# 金融工程

# 第十三章 期权的交易策略及其应用

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http:// efinance.org.cn




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# 目录

期权交易头寸及其应用（自学）

期权交易策略及其应用

期权组合盈亏图的算法

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# 运用期权进行静态风险管理

* 静态风险管理是指一次交易之后直至到期都不再调整的风险管理交易
* 它是和需要不断调整头寸的动态风险管理相对而言的
* 运用期权对其他资产进行静态风险管理是通过期权多头实现的

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# 静态风险管理

* 期限
* 行权价
* 期权组合

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# 运用期权多头进行杠杆投资

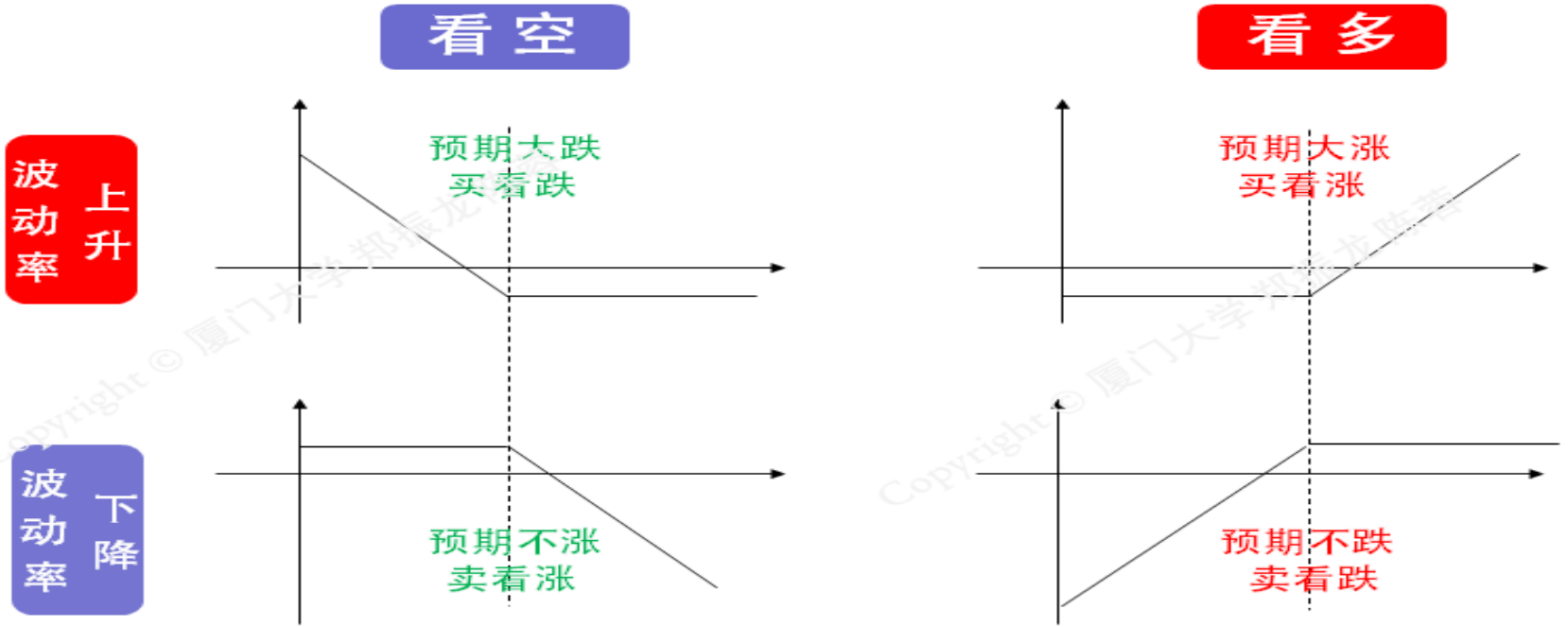
* 2020年1月23日凌晨，武汉宣布自1月23日10时封城。此消息显然属于利空消息。如果投资者看空沪深300股价指数,则其有以下两种投资途径：
* (1)按1月23日开盘价4093.0点做空2月到期的沪深300股指期货IF2002。
* (2)按1月23日开盘价1.4点的价格买进2月到期、行权价为3600点的沪深300股指看跌期权。

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* 1月24日到2月2日，中国股市因为春节假期休市。受疫情影响，2月3日股市开市后，中国股市大跌。
* 到2月3日收盘，IF2002跌到3591.2点，2个交易日期货价格下跌12.26%。按股指期货10%的保证金计算，2个交易日做空期货的收益率达到122.6%。
* 而2月到期、行权价为3600点的沪深300股指看跌期权涨到133.0点，两个交易日涨幅高达9500%！

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# 卖空期权进行投机




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# 运用期权的其他方法

* 2020年2月5日,投资者A长期看好上证50指数,但认为市场短期很有可能向下调整。
* 备兑开仓：
  * 2月5日以2.811元的价格买入1万份上证50EFT,同时以0.0521的价格卖出1份2月26日到期、行权价格为2.80元的上证50ETF看涨期权,收入期权费521元。
  * 假如2月26日到期时上证50ETF股票价格未能高于2.80元,多头不行权,则A白赚了期权费,相当于持股成本降低了1.85%。
  * 当然,如果2月26日到期时上证50ETF价格高于2.80元,A就必须以2.80元卖出手中的ETF,其实际卖出价是2.8521元,赚了1.86%。

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* 做空看跌期权
  * 2月5日以0.0513元的价格卖出1份行权价格为2.80元、2月26日到期的50ETF看跌期权，收入期权费513元。
  * 如果到期时上证50ETF未跌穿2.80元，则A净赚513元期权费。
  * 如果到期时上证50ETF跌穿2.80元，则对于A来说，他实际上以2.80-0.0513=2.7487元的价格买入了50ETF，比2月5日直接购买上证50ETF节省了1.83%的成本。

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期权交易策略及其应用

期权组合盈亏图的算法

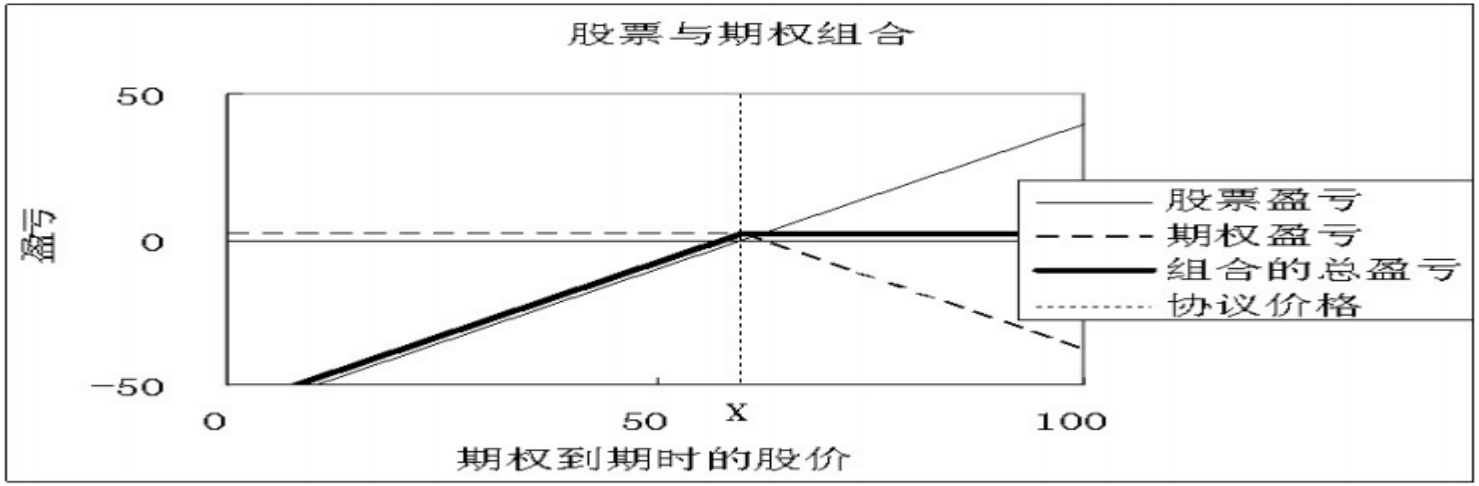
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# 期权组合

* 根据各自对未来标的资产现货价格概率分布的不同预期，以及自己的风险收益偏好，进行期权组合
* 包括标的资产与期权的组合、期权与期权的组合。
* 只要期权行权价格足够多，组合种类是无限的。
* 分析方法：
  * Payoff和profit分布图
  * 盈亏状况分析表

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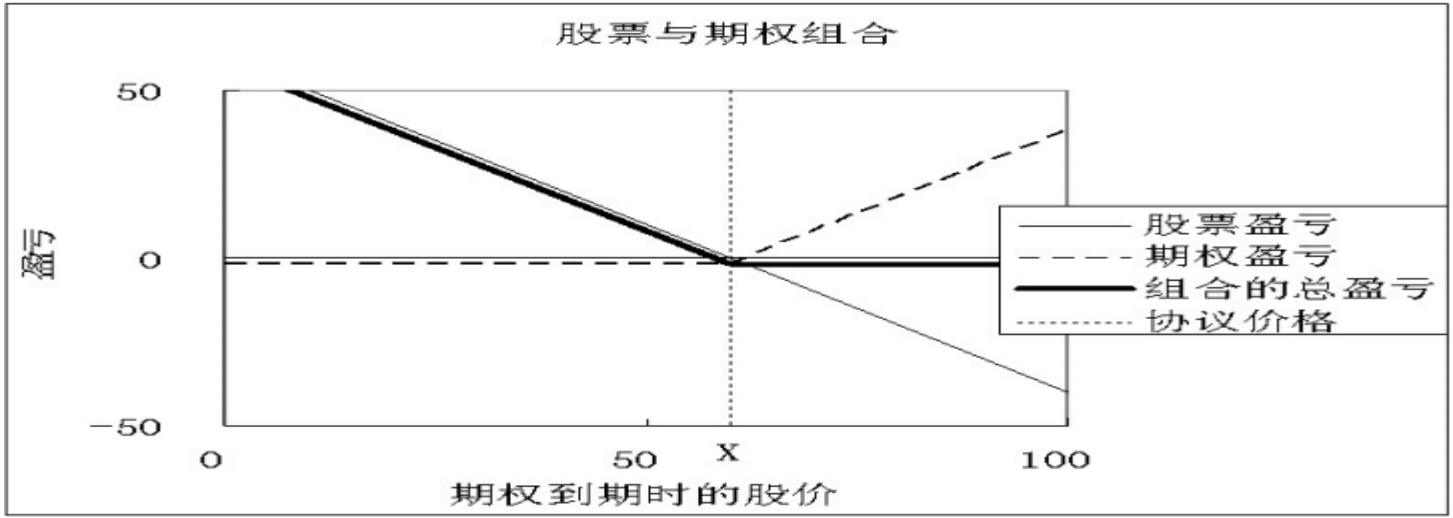
# 标的资产与期权的组合




(a)标的资产多头与看涨期权空头的组合

(有担保的看涨期权空头)

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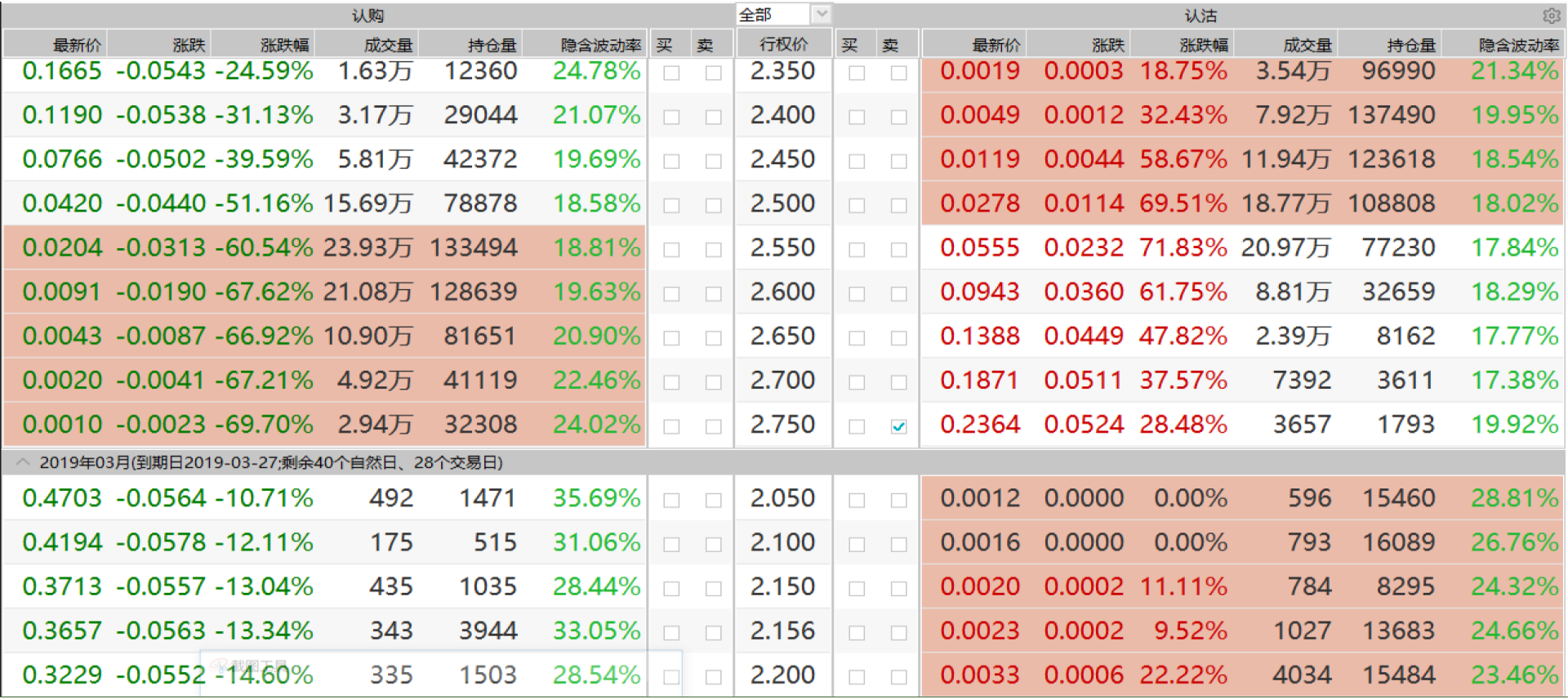

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# 期权与期权的组合

* 期权分类
  * 期权种类：看涨/看跌
  * 期权合约的具体规格：行权价格/期限/多空
* 期权组合的种类
  * 行权价格——差价组合（期限相同）
  * 期限——差期组合（行权价格相同）
  * 行权价格和期限同时不同——对角组合（同种期权）
  * 不同种类（call/put）——混合期权
  * 多头/空头——牛市/熊市、正向/反向、顶部/底部等

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# 可用的工具



| 认购 | | | | | | | | 全部 | | | 认沽 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 最新价 | 涨跌 | 涨跌幅 | 成交量 | 持仓量 | 隐含波动率 | 买 | 卖 | 行权价 | 买 | 卖 | 最新价 | 涨跌 | 涨跌幅 | 成交量 | 持仓量 | 隐含波动率 |
| 0.1665 | -0.0543 | -24.59% | 1.63万 | 12360 | 24.78% | | | 2.350 | | | 0.0019 | 0.0003 | 18.75% | 3.54万 | 96990 | 21.34% |
| 0.1190 | -0.0538 | -31.13% | 3.17万 | 29044 | 21.07% | | | 2.400 | | | 0.0049 | 0.0012 | 32.43% | 7.92万 | 137490 | 19.95% |
| 0.0766 | -0.0502 | -39.59% | 5.81万 | 42372 | 19.69% | | | 2.450 | | | 0.0119 | 0.0044 | 58.67% | 11.94万 | 123618 | 18.54% |
| 0.0420 | -0.0440 | -51.16% | 15.69万 | 78878 | 18.58% | | | 2.500 | | | 0.0278 | 0.0114 | 69.51% | 18.77万 | 108808 | 18.02% |
| 0.0204 | -0.0313 | -60.54% | 23.93万 | 133494 | 18.81% | | | 2.550 | | | 0.0555 | 0.0232 | 71.83% | 20.97万 | 77230 | 17.84% |
| 0.0091 | -0.0190 | -67.62% | 21.08万 | 128639 | 19.63% | | | 2.600 | | | 0.0943 | 0.0360 | 61.75% | 8.81万 | 32659 | 18.29% |
| 0.0043 | -0.0087 | -66.92% | 10.90万 | 81651 | 20.90% | | | 2.650 | | | 0.1388 | 0.0449 | 47.82% | 2.39万 | 8162 | 17.77% |
| 0.0020 | -0.0041 | -67.21% | 4.92万 | 41119 | 22.46% | | | 2.700 | | | 0.1871 | 0.0511 | 37.57% | 7392 | 3611 | 17.38% |
| 0.0010 | -0.0023 | -69.70% | 2.94万 | 32308 | 24.02% | | | 2.750 | | ☑ | 0.2364 | 0.0524 | 28.48% | 3657 | 1793 | 19.92% |
| 2019年03月(到期日2019-03-27;剩余40个自然日、28个交易日) | | | | | | | | | | | | | | | | |
| 0.4703 | -0.0564 | -10.71% | 492 | 1471 | 35.69% | | | 2.050 | | | 0.0012 | 0.0000 | 0.00% | 596 | 15460 | 28.81% |
| 0.4194 | -0.0578 | -12.11% | 175 | 515 | 31.06% | | | 2.100 | | | 0.0016 | 0.0000 | 0.00% | 793 | 16089 | 26.76% |
| 0.3713 | -0.0557 | -13.04% | 435 | 1035 | 28.44% | | | 2.150 | | | 0.0020 | 0.0002 | 11.11% | 784 | 8295 | 24.32% |
| 0.3657 | -0.0563 | -13.34% | 343 | 3944 | 33.05% | | | 2.156 | | | 0.0023 | 0.0002 | 9.52% | 1027 | 13683 | 24.66% |
| 0.3229 | -0.0552 | -14.60% | 335 | 1503 | 28.54% | | | 2.200 | | | 0.0033 | 0.0006 | 22.22% | 4034 | 15484 | 23.46% |

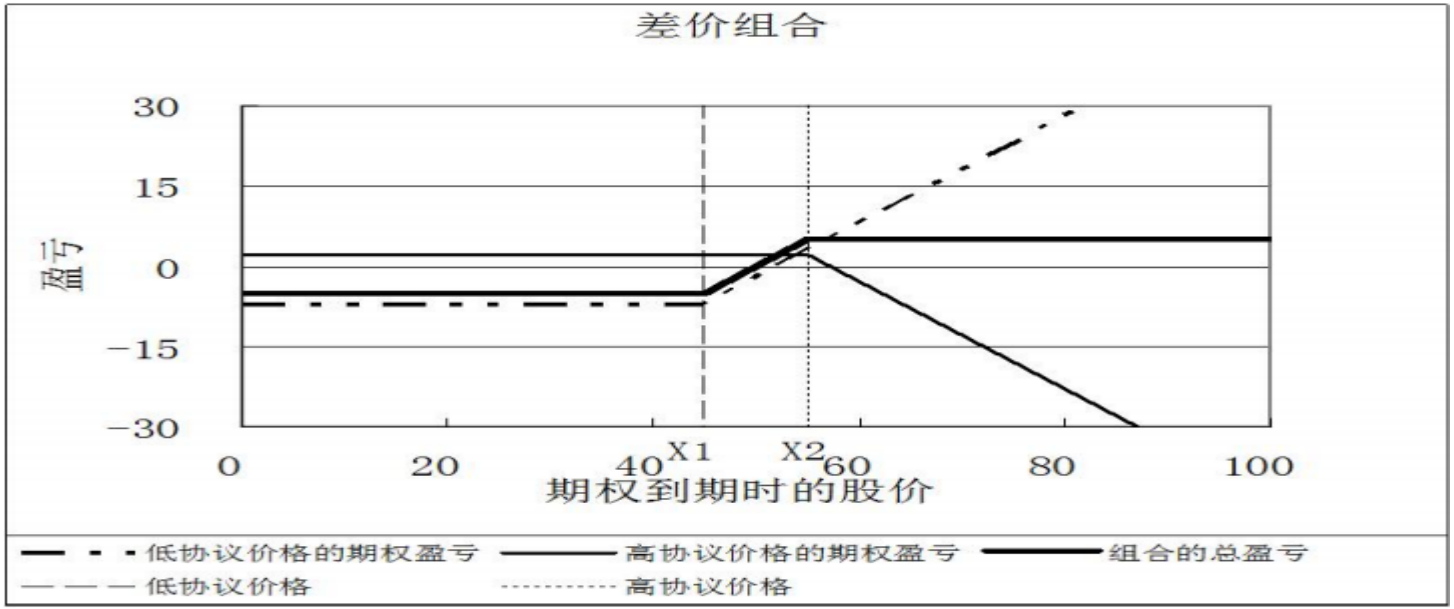
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# 差价组合

- 持有相同期限、不同行权价格的两个或多个同种期权头寸组合（即同是看涨期权，或者同是看跌期权）。
- 其主要类型
  - 牛市差价组合（买低卖高）
  - 熊市差价组合（买高卖低）
  - 蝶式差价组合等

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# 牛市差价（Bull Spreads）：看涨



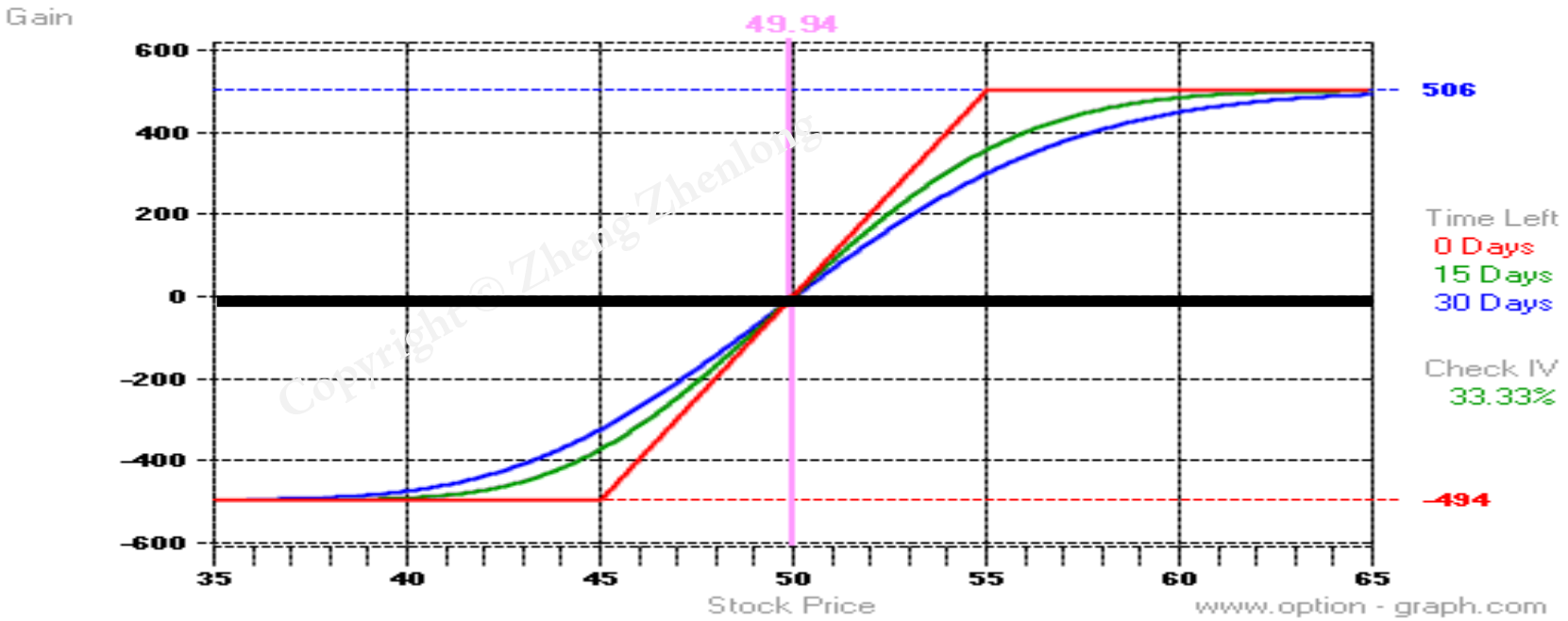

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# 盈亏均衡

**Position Details : Bull Call to compare to Modified Bullish Butterfly**

| | | |
|---|---|---|
| Stock at 50, 1 Short Call, 55 strike, 30 days, IV 33.33%, Option Price $0.45 = | $45.00 | credit |
| Stock at 50, 1 Long Call, 45 strike, 30 days, IV 33.33%, Option Price $5.39 = | $539.00 | debit |

Total Position = $494.00 DEBIT



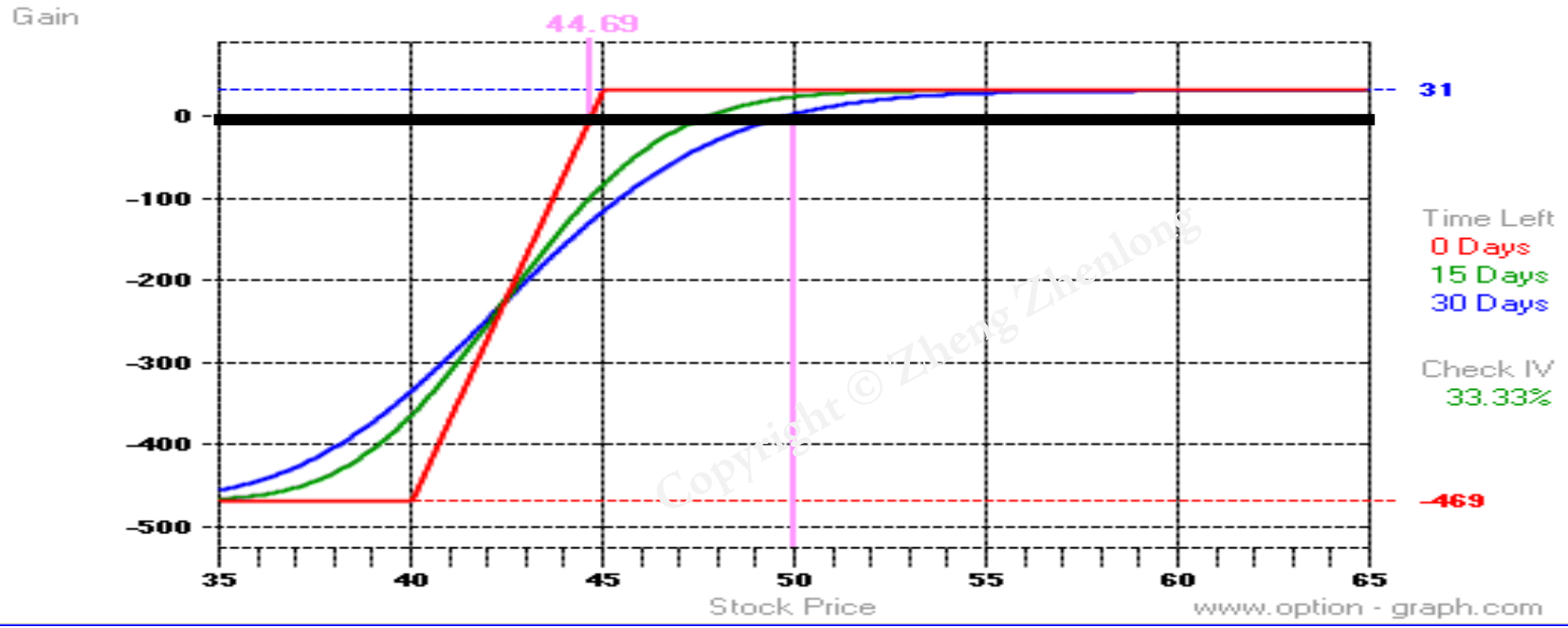

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# 大概率中小奖

**Position Details : Bull Call (2 strikes down to 1 strike down)**

| | | |
|---|---|---|
| Stock at 50, 1 Long Call, 40 strike, 30 days, IV 33.33%, Option Price $10.08 = | $1,008.00 | debit |
| Stock at 50, 1 Short Call, 45 strike, 30 days, IV 33.33%, Option Price $5.39 = | $539.00 | credit |

Total Position = $469.00 DEBIT



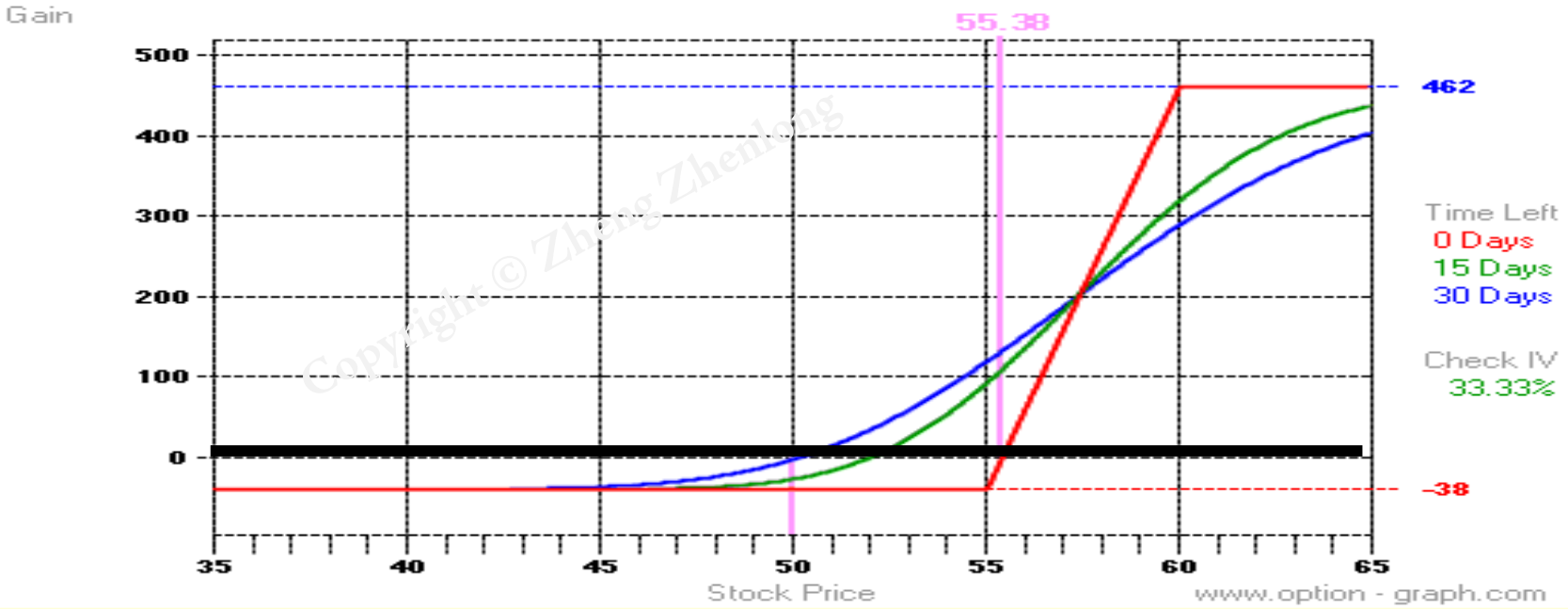

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# 小概率中大奖

**Position Details : Bull Call (1 strike up to 2 strikes up)**

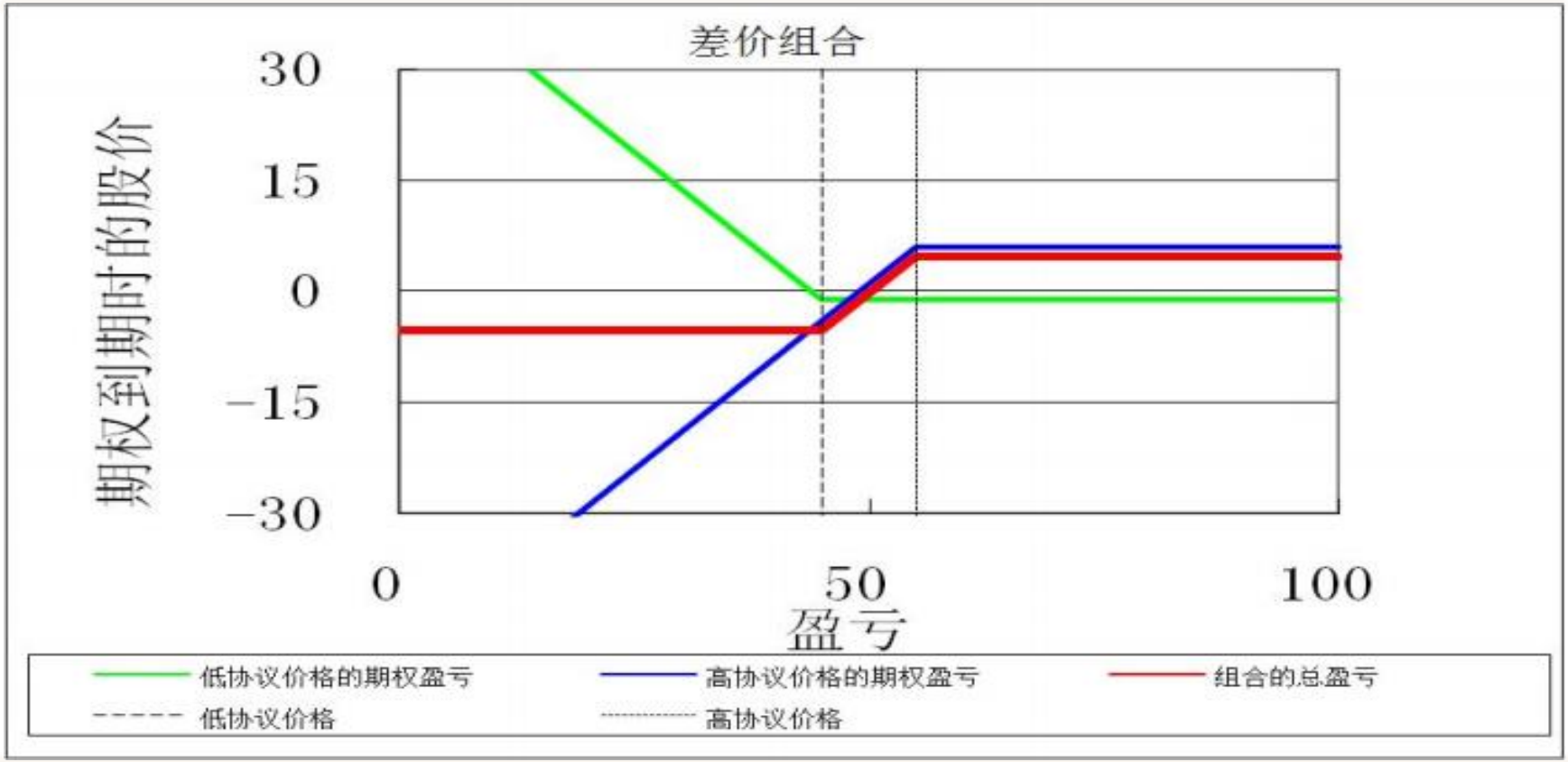
| | | |
|---|---|---|
| Stock at 50, 1 Long Call, 55 strike, 30 days, IV 33.33%, Option Price $0.45 = | $45.00 | debit |
| Stock at 50, 1 Short Call, 60 strike, 30 days, IV 33.33%, Option Price $0.07 = | $7.00 | credit |

Total Position = $38.00 DEBIT




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# 牛市差价：看跌



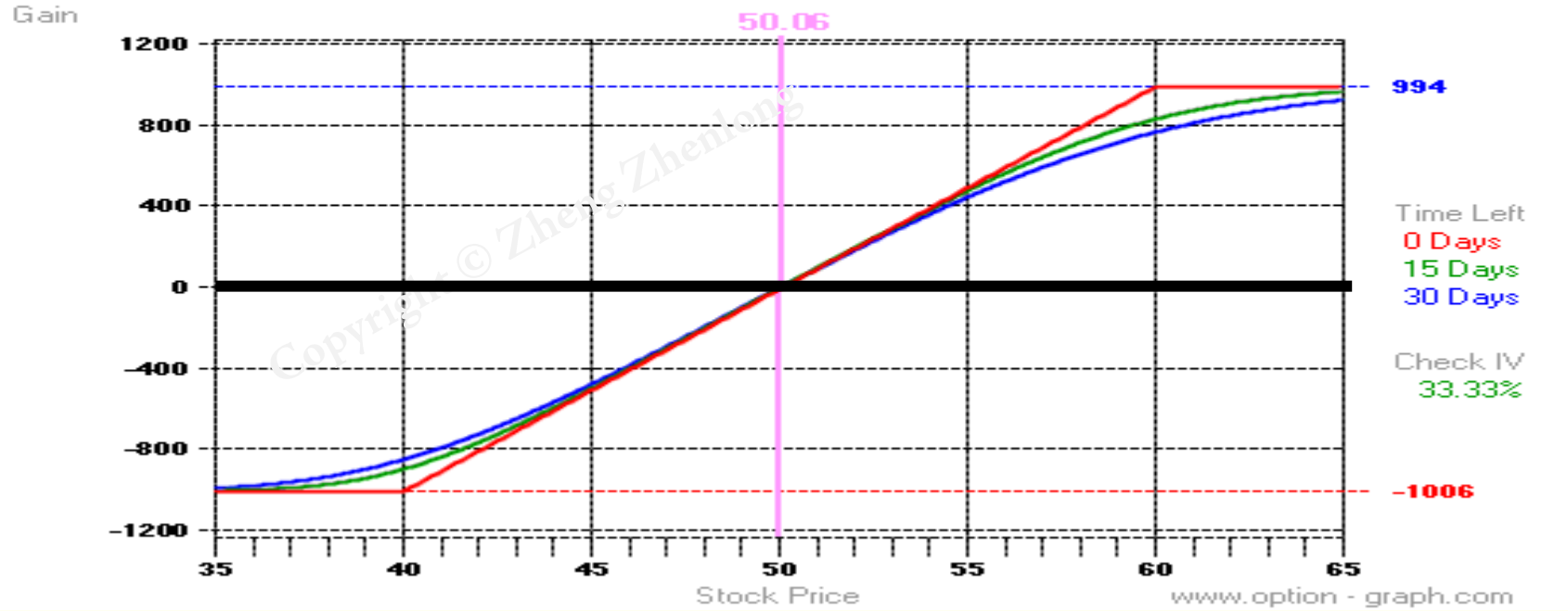

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# 盈亏均衡

**Position Details : Bull Put (2 strikes down to 2 strikes up)**

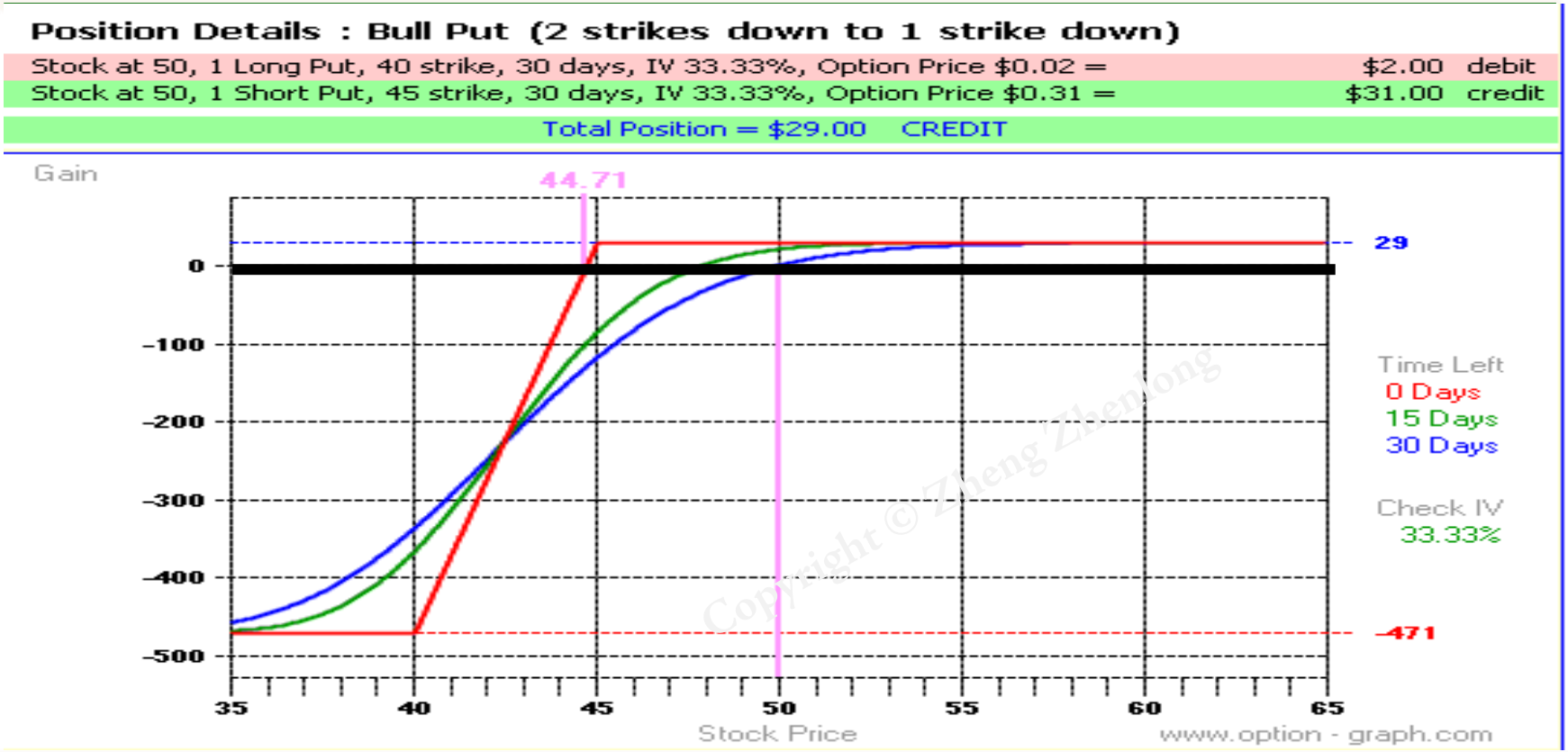
| | | |
|---|---|---|
| Stock at 50, 1 Long Put, 40 strike, 30 days, IV 33.33%, Option Price $0.02 = | $2.00 | debit |
| Stock at 50, 1 Short Put, 60 strike, 30 days, IV 33.33%, Option Price $9.96 = | $996.00 | credit |

Total Position = $994.00 CREDIT




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# 大概率中小奖



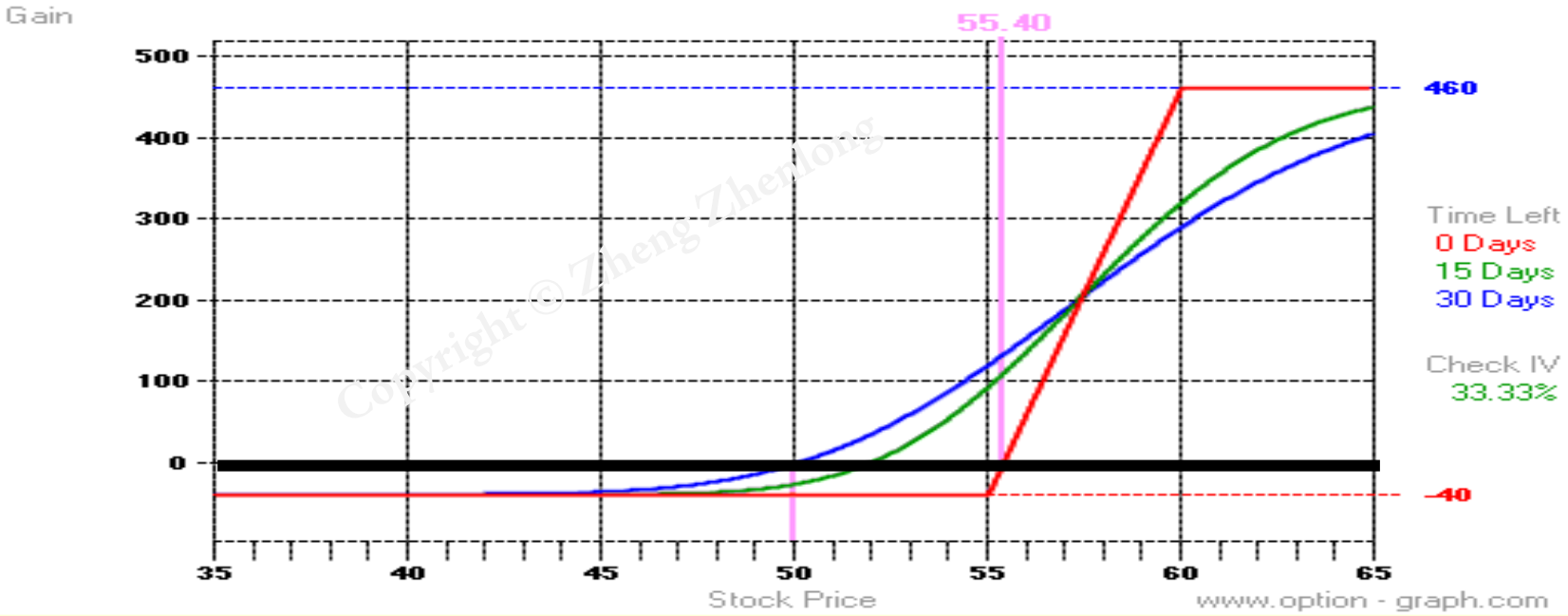

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# 小概率中大奖

**Position Details : Bull Put (1 strike up to 2 strikes up)**

| | | |
|---|---|---|
| Stock at 50, 1 Long Put, 55 strike, 30 days, IV 33.33%, Option Price $5.36 = | $536.00 | debit |
| Stock at 50, 1 Short Put, 60 strike, 30 days, IV 33.33%, Option Price $9.96 = | $996.00 | credit |

Total Position = $460.00 CREDIT




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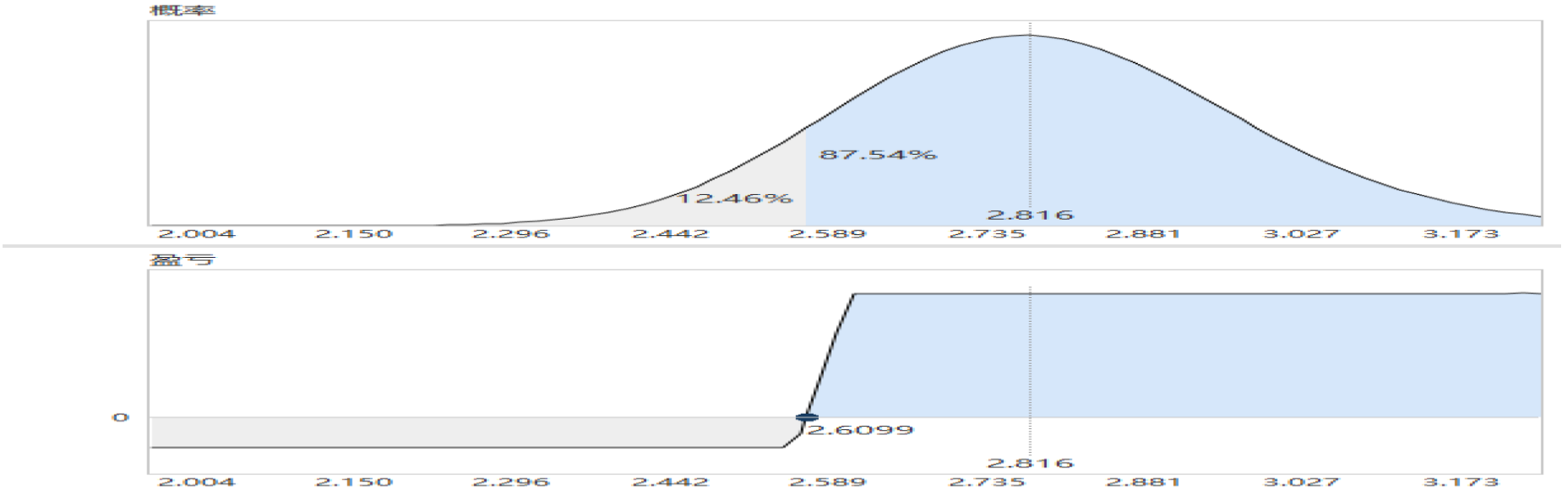
# 牛市差价组合（19.2.25）

| 删除 | 分析 | 买卖 | 名称 | 手数 | 价格类型 | 价格 | 买价 | 卖价 | 隐含波动率 | Delta | Gamma | Vega | Theta | 合约乘数 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| × | ☑ | 卖 | 50ETF购3月2.65 | 1 | 买价 | 0.2500 | 0.2500 | 0.2501 | 47.00% | 0.9090 | 1.2147 | 0.0013 | -0.0005 | 10000 |
| × | ☑ | 买 | 50ETF购3月2.60 | 1 | 卖价 | 0.2599 | 0.2595 | 0.2599 | 37.38% | 0.9584 | 0.6599 | 0.0007 | -0.0004 | 10000 |

⊞加入标的　⊞保存策略

最大收益: 401.00　最大亏损: -99.00

总Delta: 605.93　总Gamma: -3861.88　总Vega: -5.72　总Theta: 1.95




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# 牛市差价的适用情形

* 怎样的情况下牛市差价组合有利？（为什么要构建？）
  * 低行权价期权价格相对较低，买低卖高的概率赔率组合较有利。

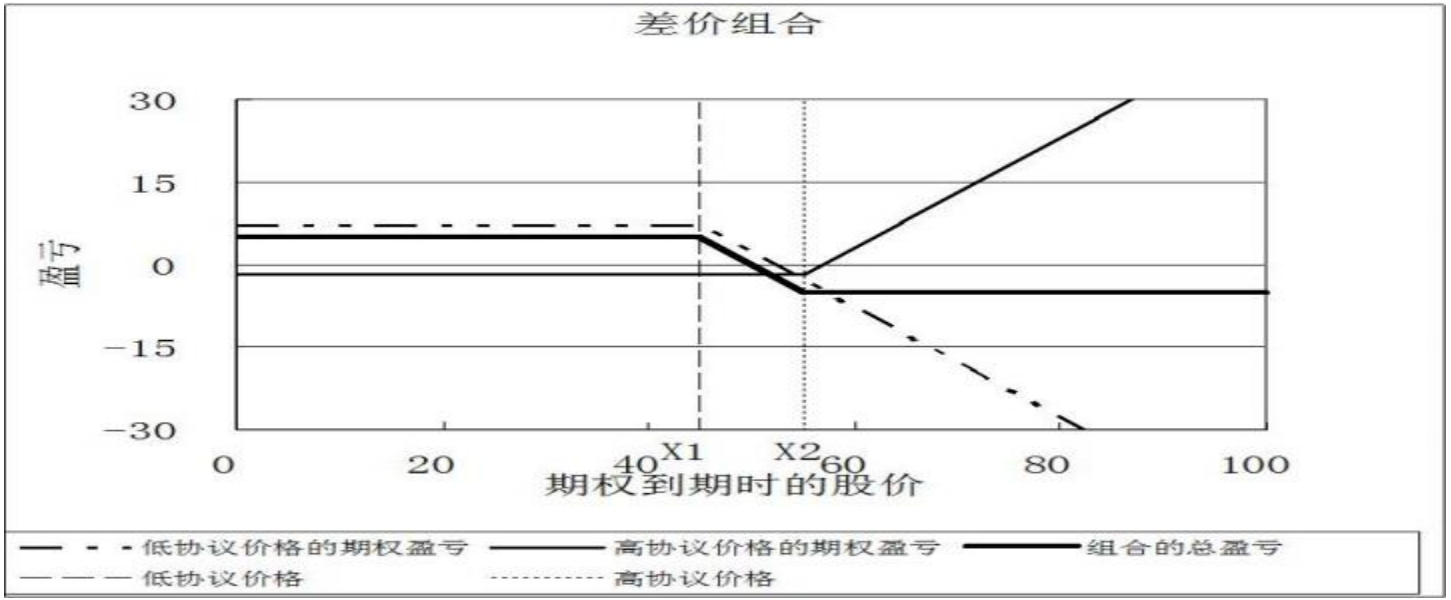
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# 牛市差价：看涨VS看跌

- 看涨期权和看跌期权的牛市差价组合
  - 看涨期权的牛市差价组合：期初现金流为负，但期末回报大于看跌期权的牛市差价组合
  - 看跌期权的牛市差价组合：期初现金流为正，但期末回报小于看涨期权的牛市差价组合

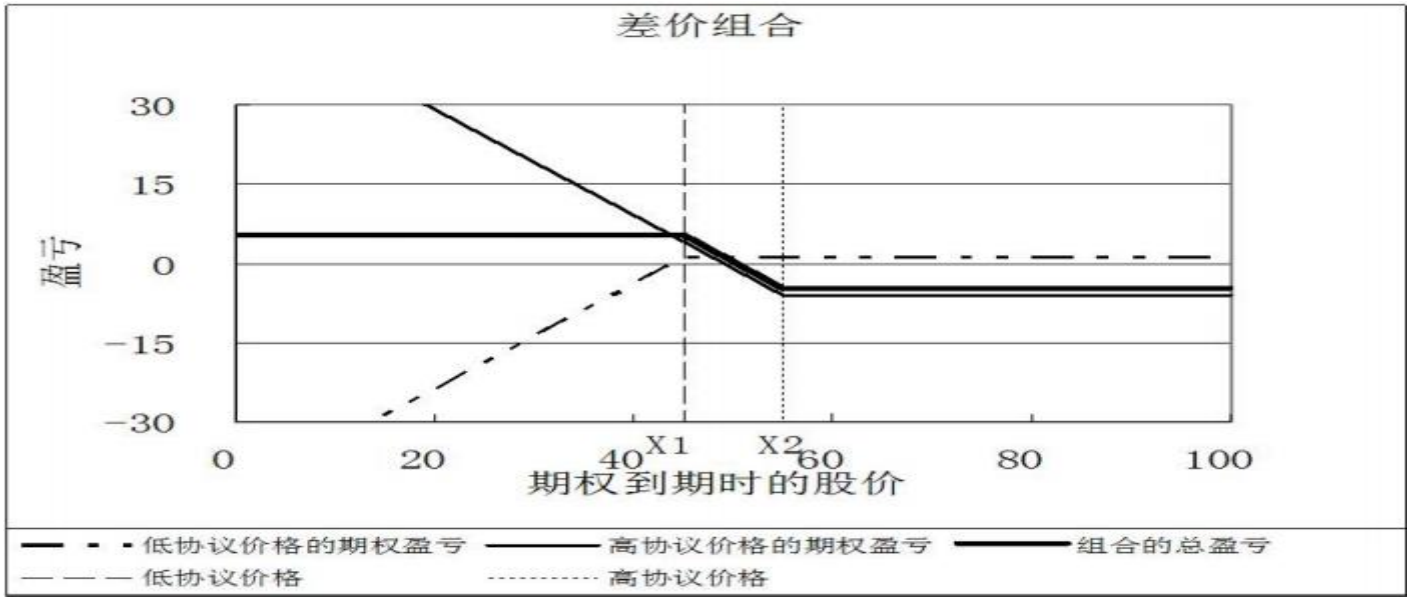
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# 熊市差价（Bear Spreads）：看涨




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# 熊市差价：看跌




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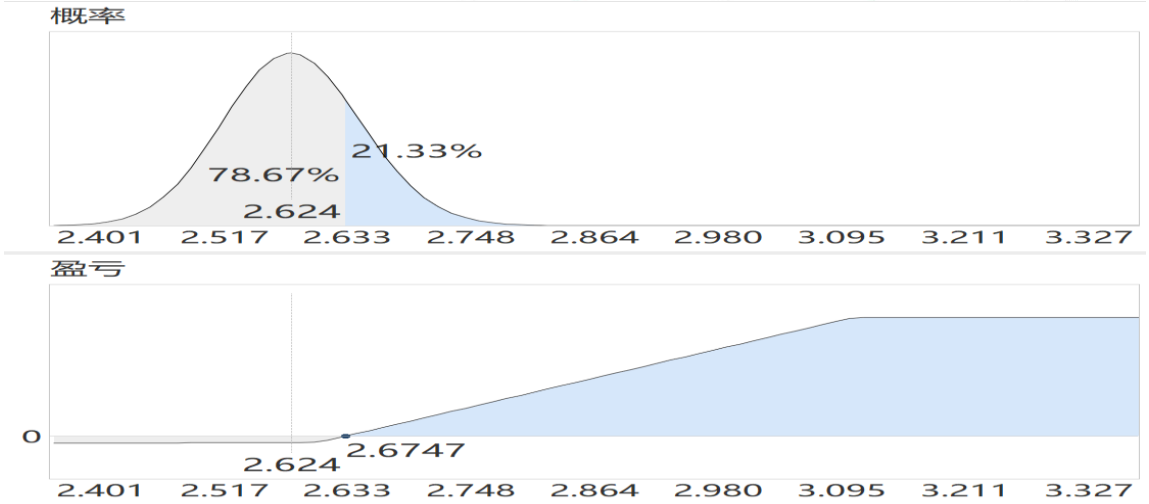
# 牛市差价组合（20.3.20）

| 删除 | 分析 | 买卖 | 名称 | 手数 | 价格类 | 价格 | 买价 | 卖价 | IV | Delta | Gamma | Vega | Theta | 合约乘数 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| × | ✓ | 买 | 50ETF沽3月 | 1 | 卖 | 0.049 | 0.0495 | 0.0496 | 26.37% | -0.6040 | 4.3190 | 0.0013 | -0.0028 | 10000 |
| × | ✓ | 卖 | 50ETF沽3月 | 1 | 买 | 0.523 | 0.5231 | 0.5297 | 71.88% | -1.0000 | - | - | 0.0002 | 10163 |

⊞加入标的　⊞保存策略

最大收益: 4,820.27　最大亏损: -683.02

总Delta: 3,669.2066　总Gamma: 57,621.7325　总Vega: 8.8171　总Theta: -40.4856




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# 熊市差价的适用情形

* 怎样的情况下熊市差价组合有利？（为什么要构建？）
  * 高行权价期权价格相对较低，买高卖低的概率赔率组合较有利。

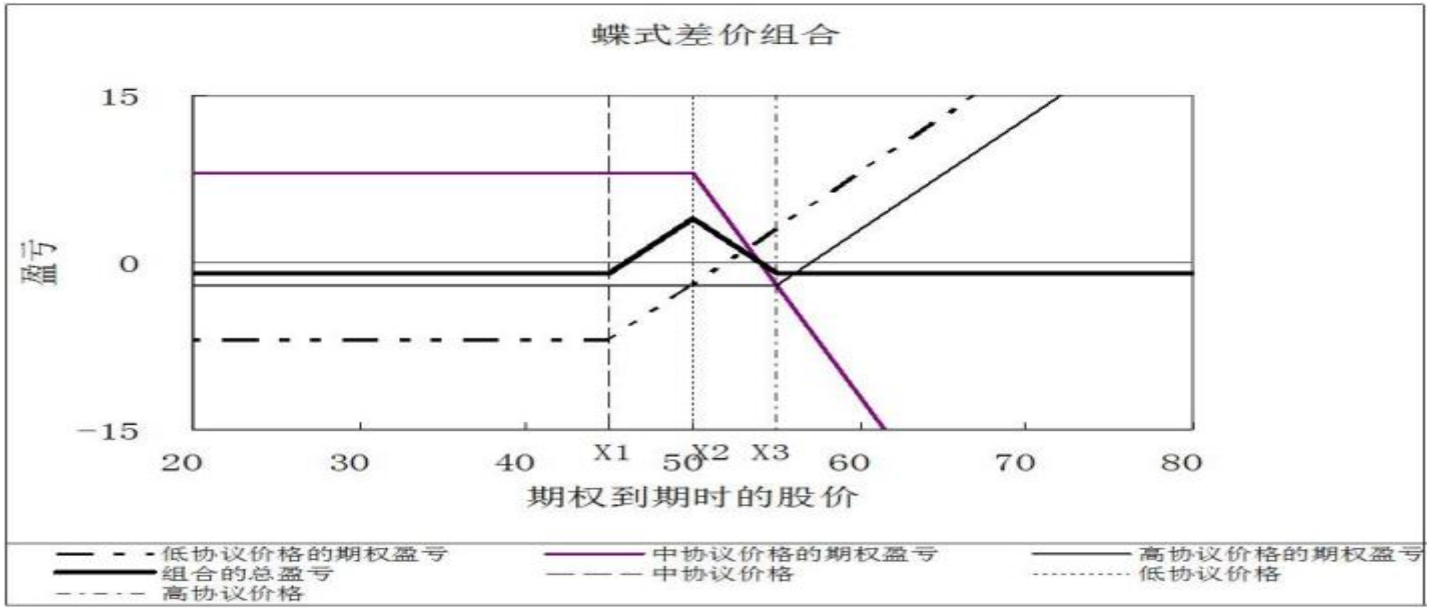
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# 熊市差价：看涨VS看跌

* 看涨期权的熊市差价组合和看跌期权的熊市差价组合的差别在于，前者在期初有正的现金流，后者在期初则有负的现金流，但后者的期末回报大于前者。
* 熊市差价组合刚好跟牛市差价组合相反，两者的图形刚好以X轴对称。

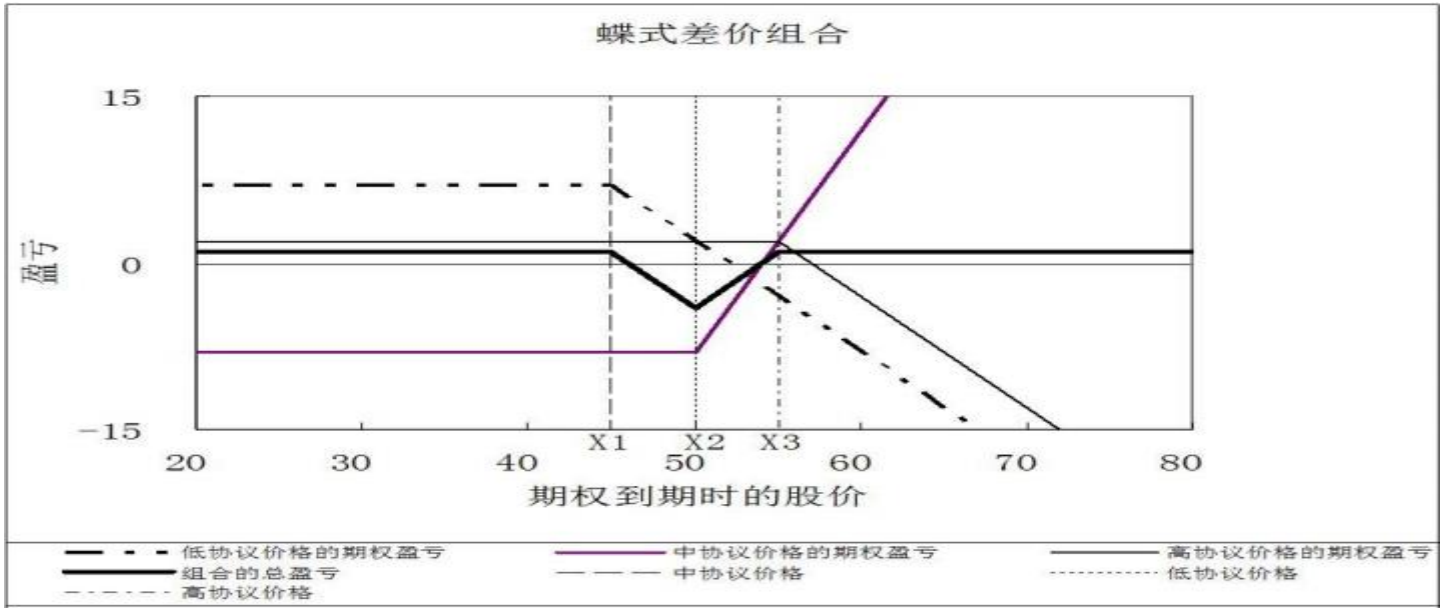
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# 蝶式差价（Butterfly Spreads）：看涨正向




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# 蝶式差价：看涨反向




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* 蝶式差价组合由四份具有相同期限、不同行权价格的同种期权头寸组成。
* 若 $K_1 < K_2 < K_3$，且 $K_2 = (K_1 + K_3)/2$，则蝶式差价组合有如下四种：

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* 看涨期权的正向蝶式差价组合：由行权价格分别为 $K_1$ 和 $K_3$ 的看涨期权多头和两份行权价格为 $K_2$ 的看涨期权空头组成
* 看涨期权的反向蝶式差价组合：由行权价格分别为 $K_1$ 和 $K_3$ 的看涨期权空头和两份行权价格为 $K_2$ 的看涨期权多头组成

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* 看跌期权的正向蝶式差价组合：由行权价格分别为 $K_1$ 和 $K_3$ 的看跌期权多头和两份行权价格为 $K_2$ 的看跌期权空头组成
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* 若$K_2 = (K_1 + K_3)/2$，无论用认购还是看跌期权组合，正向和反向结果都相同，并且初始投资也相同。（如何证明？PCP）

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# 蝶式差价适用情形

* 为何构建蝶式差价组合：预期价格会在一定的区间内波动

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# 看涨期权正向蝶式差价组合的盈亏状况表

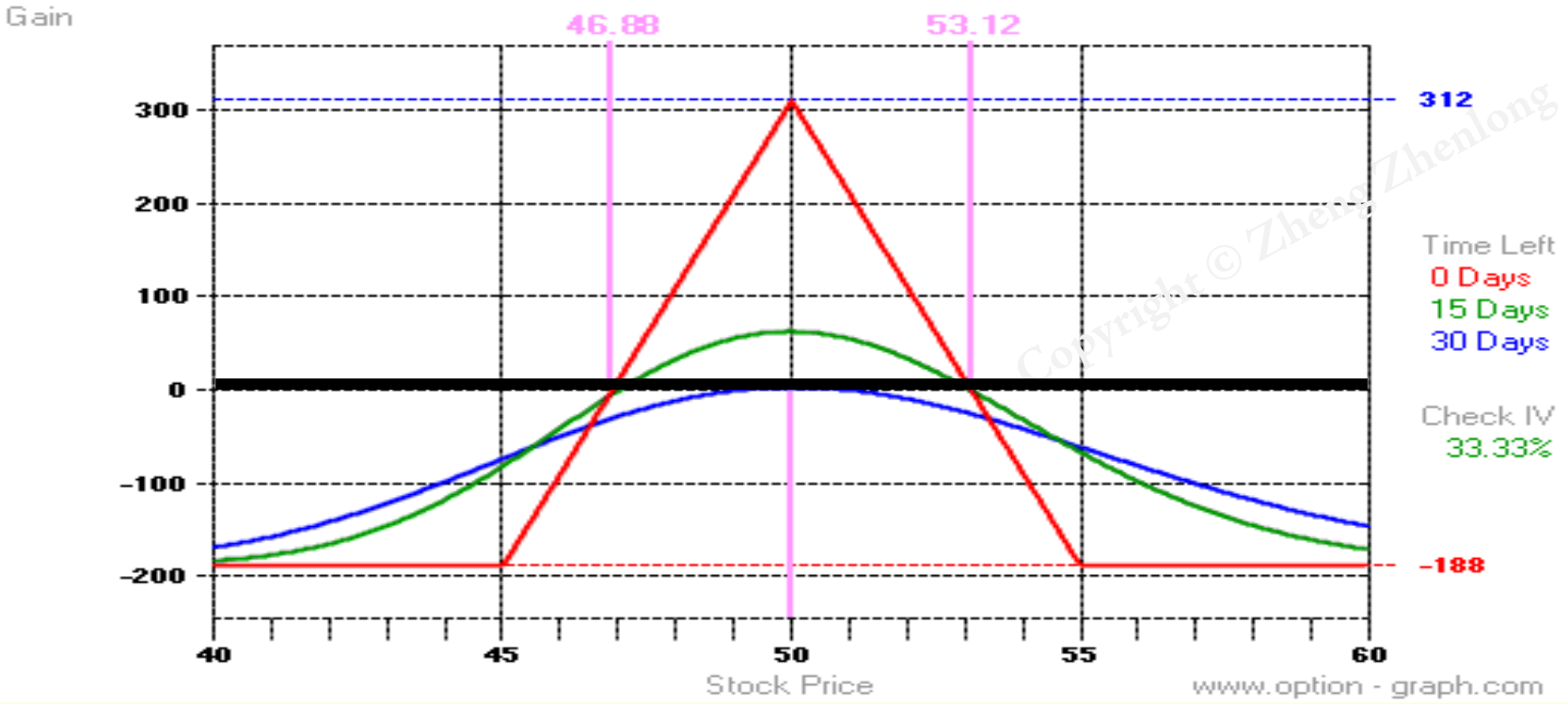
| $S_T$的范围 | $c_1$盈亏 | 2份$c_2$盈亏 | $c_3$的盈亏 | 总盈亏 |
|---|---|---|---|---|
| $S_T \le K_1$ | $0-c_1$ | $0+2c_2$ | $0-c_3$ | $2c_2-c_1-c_3$ |
| $K_1 < S_T \le K_2$ | $S_T-K_1-c_1$ | $0+2c_2$ | $0-c_3$ | $S_T-K_1+2c_2-c_1-c_3$ |
| $K_2 < S_T \le K_3$ | $S_T-K_1-c_1$ | $2K_2-2S_T+2c_2$ | $0-c_3$ | $K_3-S_T+2c_2-c_1-c_3$ |
| $S_T > K_3$ | $S_T-K_1-c_1$ | $2K_2-2S_T+2c_2$ | $S_T-K_3$ | $2c_2-c_1-c_3$ |

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## Position Details : Butterfly with all Calls

| | | |
|---|---|---|
| Stock at 50, 1 Short Call, 50 strike, 30 days, IV 33.33%, Option Price $1.98 = | $198.00 | credit |
| Stock at 50, 1 Short Call, 50 strike, 30 days, IV 33.33%, Option Price $1.98 = | $198.00 | credit |
| Stock at 50, 1 Long Call, 55 strike, 30 days, IV 33.33%, Option Price $0.45 = | $45.00 | debit |
| Stock at 50, 1 Long Call, 45 strike, 30 days, IV 33.33%, Option Price $5.39 = | $539.00 | debit |

Total Position = $188.00 DEBIT



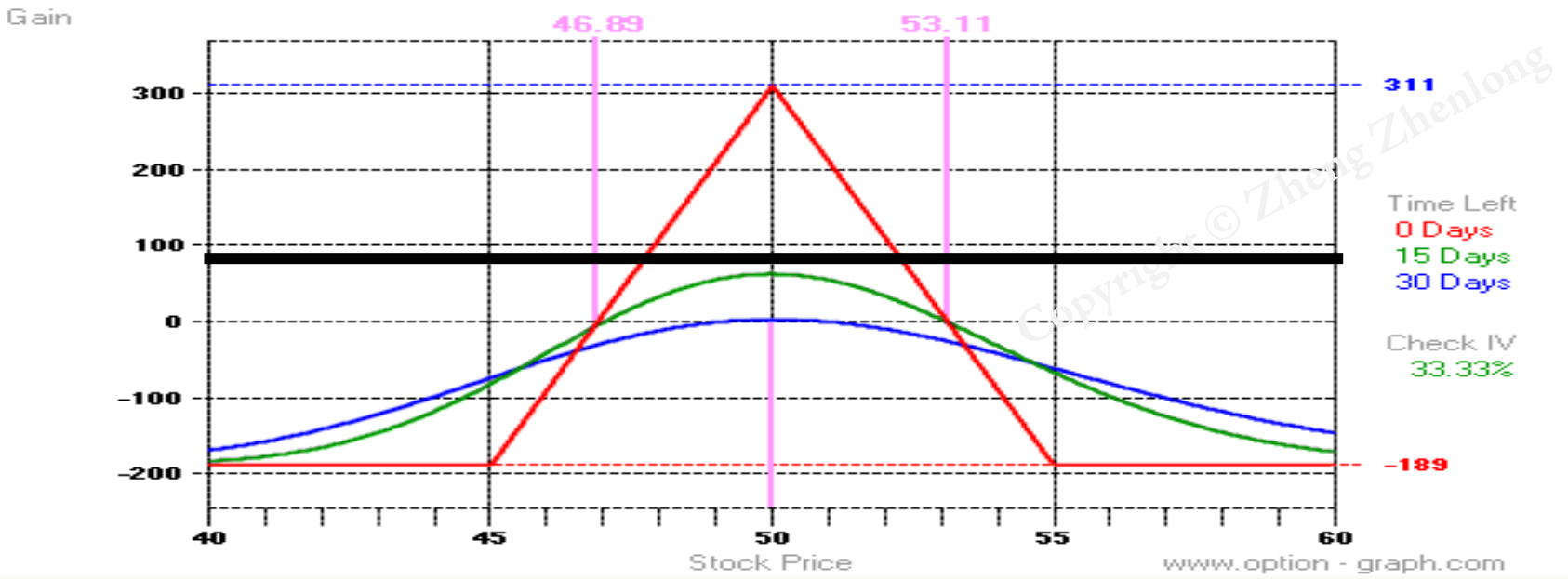

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# 看跌期权的正向蝶式差价组合

**Position Details : Butterfly with all Puts**

| | | |
|---|---|---|
| Stock at 50, 1 Short Put, 50 strike, 30 days, IV 33.33%, Option Price $1.89 = | $189.00 | credit |
| Stock at 50, 1 Short Put, 50 strike, 30 days, IV 33.33%, Option Price $1.89 = | $189.00 | credit |
| Stock at 50, 1 Long Put, 55 strike, 30 days, IV 33.33%, Option Price $5.36 = | $536.00 | debit |
| Stock at 50, 1 Long Put, 45 strike, 30 days, IV 33.33%, Option Price $0.31 = | $31.00 | debit |

Total Position = $189.00 DEBIT




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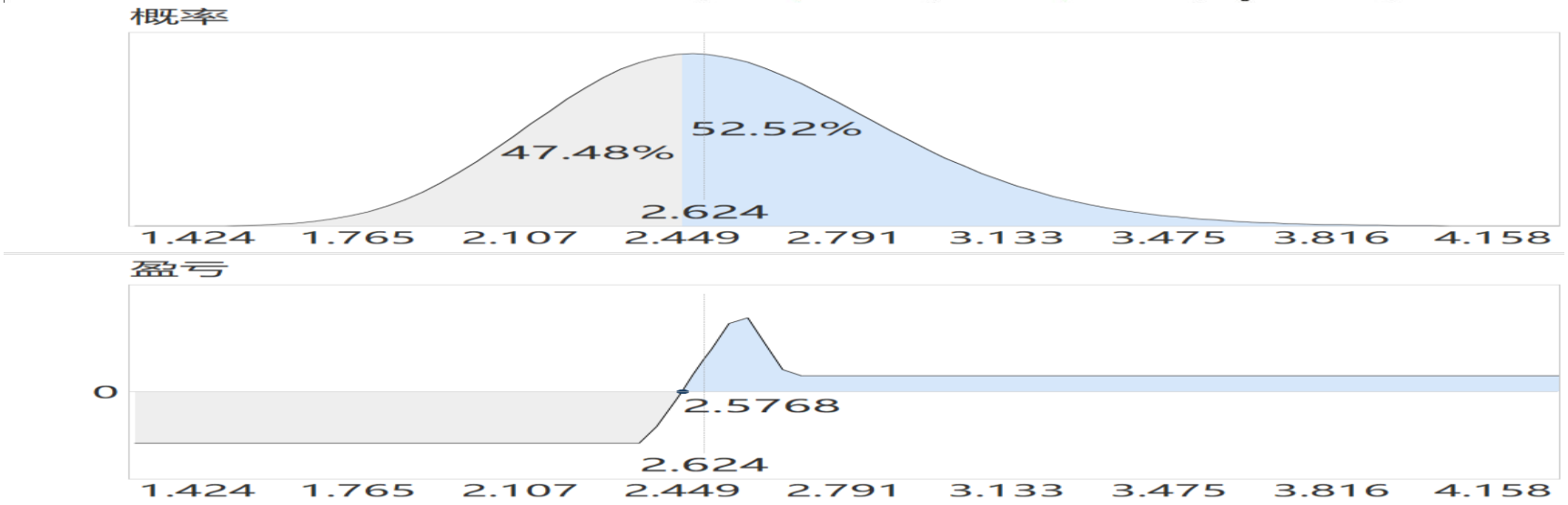
# 实例：2020.3.20 正向蝶式差价组合

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 买 | 50ETF购6月 | 1 | 卖 | 0.227 | 0.2213 | 0.2276 | 27.81% | 0.6771 | 1.0007 | 0.0049 | -0.0007 | 10000 |
| 买 | 50ETF购6月 | 1 | 卖 | 0.089 | 0.0880 | 0.0894 | 28.16% | 0.3559 | 1.0389 | 0.0050 | -0.0007 | 10000 |
| 卖 | 50ETF购6月 | 2 | 买 | 0.120 | 0.1201 | 0.1261 | 27.90% | 0.4589 | 1.1063 | 0.0054 | -0.0008 | 10000 |

加入标的　保存策略

最大收益: 1,232.00　最大亏损: -768.00

总Delta: 1,150.0550　总Gamma: -1,707.7149　总Vega: -8.1877　总Theta: 1.0563



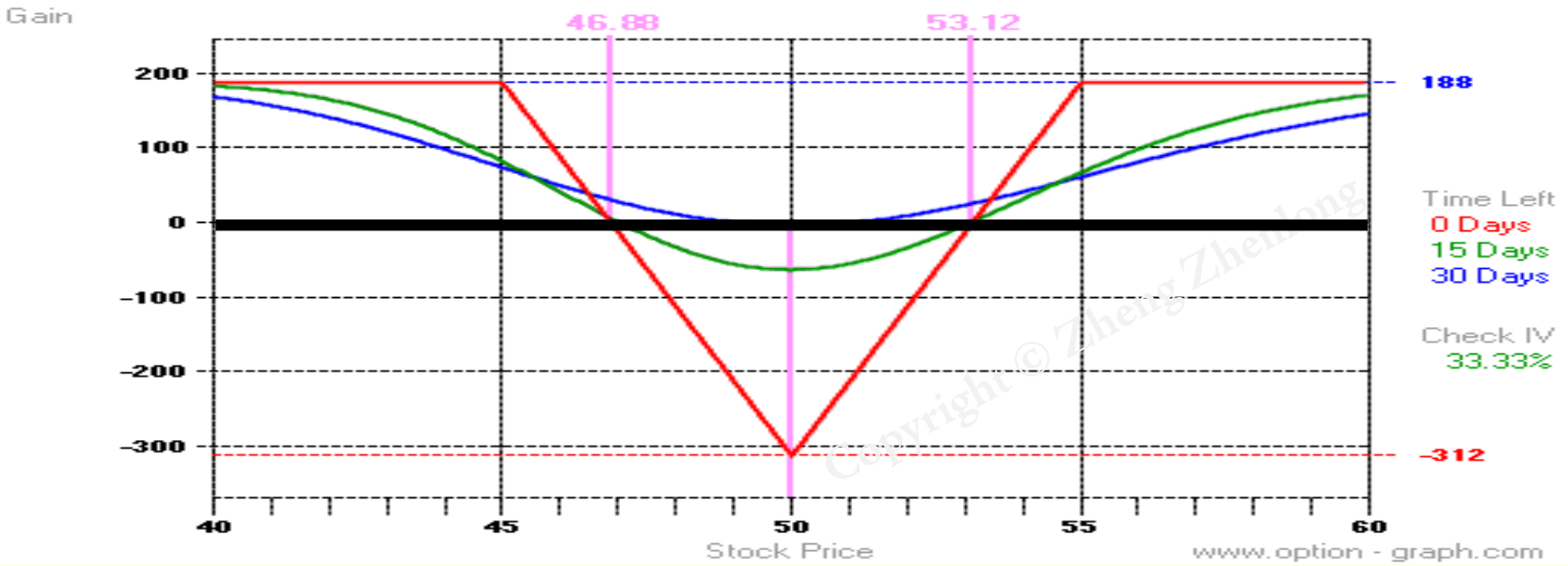

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# 看涨期权的反向蝶式差价组合

**Position Details : Reverse Butterfly with all Calls**

| | | |
|---|---|---|
| Stock at 50, 1 Long Call, 50 strike, 30 days, IV 33.33%, Option Price $1.98 = | $198.00 | debit |
| Stock at 50, 1 Long Call, 50 strike, 30 days, IV 33.33%, Option Price $1.98 = | $198.00 | debit |
| Stock at 50, 1 Short Call, 55 strike, 30 days, IV 33.33%, Option Price $0.45 = | $45.00 | credit |
| Stock at 50, 1 Short Call, 45 strike, 30 days, IV 33.33%, Option Price $5.39 = | $539.00 | credit |

Total Position = $188.00 CREDIT



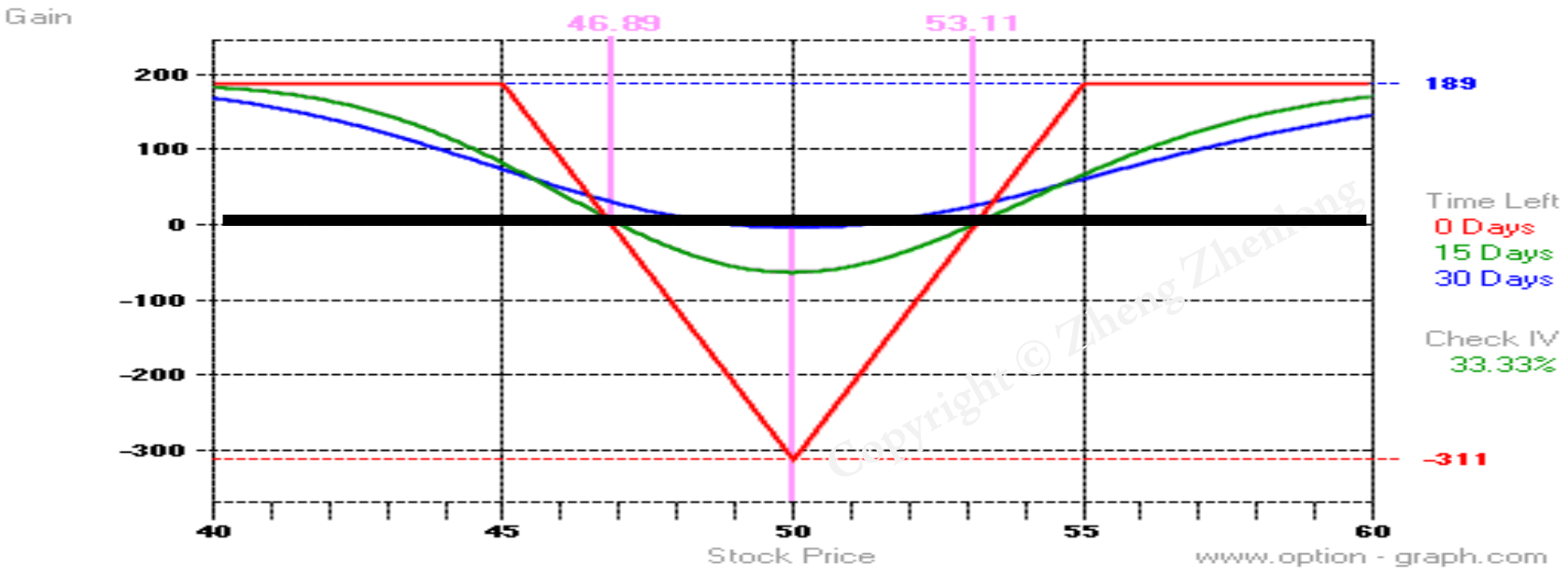

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# 看跌期权的反向蝶式差价组合

**Position Details : Reverse Butterfly with all Puts**

| | | |
|---|---|---|
| Stock at 50, 1 Long Put, 50 strike, 30 days, IV 33.33%, Option Price $1.89 = | $189.00 | debit |
| Stock at 50, 1 Long Put, 50 strike, 30 days, IV 33.33%, Option Price $1.89 = | $189.00 | debit |
| Stock at 50, 1 Short Put, 55 strike, 30 days, IV 33.33%, Option Price $5.36 = | $536.00 | credit |
| Stock at 50, 1 Short Put, 45 strike, 30 days, IV 33.33%, Option Price $0.31 = | $31.00 | credit |

Total Position = $189.00 CREDIT




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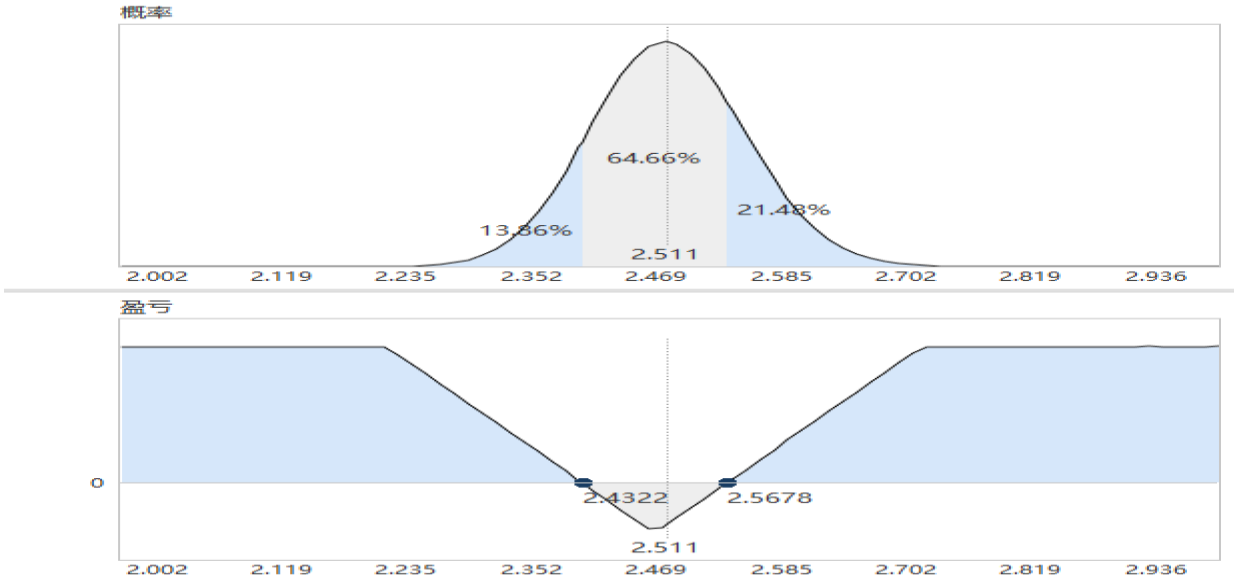
# 反向蝶式（19.2.15）

| 买卖 | 名称 | 手数 | 价格类型 | 价格 | 买价 | 卖价 | 隐含波动率 | Delta | Gamma | Vega | Theta | 合约乘数 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 卖 | 50ETF购2月2.75 | 1 | 买价 | 0.0010 | 0.0010 | 0.0011 | 24.02% | 0.0011 | 0.0505 | - | - | 10000 |
| 买 | 50ETF购2月2.50 | 2 | 卖价 | 0.0420 | 0.0419 | 0.0420 | 18.58% | 0.5764 | 5.3194 | 0.0019 | -0.0012 | 10000 |
| 卖 | 50ETF购2月2.25 | 1 | 买价 | 0.2652 | 0.2652 | 0.2654 | 33.50% | 0.9999 | 0.0042 | - | -0.0001 | 10000 |

标的　　保存策略

最大收益: 1822.00　最大亏损: -678.00

总Delta: 1518.32　总Gamma: 107396.81　总Vega: 35.50　总Theta: -23.95




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# 差期组合（Calendar Spreads）

* 差期组合：两份相同行权价格、不同期限的同种期权的不同头寸组成
* 差期组合的四种类型
  * 正向差期组合（买长卖短）
    * 一份看涨期权多头与一份期限较短的看涨期权空头的组合，称看涨期权的正向差期组合。
    * 一份看跌期权多头与一份期限较短的看跌期权空头的组合，称看跌期权的正向差期组合。

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- 反向差期组合（买短卖长）
  - 一份看涨期权多头与一份期限较长的看涨期权空头的组合，称看涨期权的反向差期组合。
  - 一份看跌期权多头与一份期限较长的看跌期权空头的组合，称看跌期权的反向差期组合。

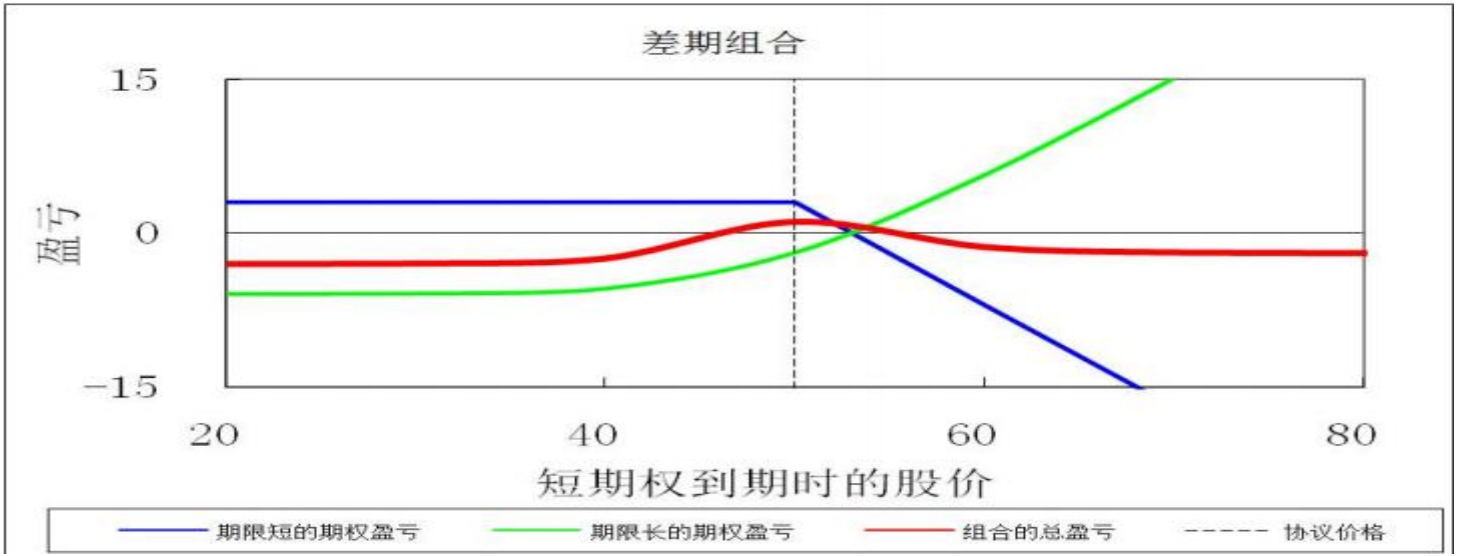
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# 看涨期权的正向差期组合的盈亏状况表

| $S_T$的范围 | 看涨期权多头的盈亏 | 看涨期权空头的盈亏 | 总盈亏 |
|---|---|---|---|
| $S_T \to \infty$ | 趋近$S_T - Ke^{-r(T^*-T)} - c_1$ | $K - S_T + c_2$ | 趋近$K - Ke^{-r(T^*-T)} + c_2 - c_1$ |
| $S_T = K$ | $K - Ke^{-r(T^*-T)} + c_{1T} - c_1$ | $c_2$ | $K - Ke^{-r(T^*-T)} + c_2 - c_1 + c_{1T}$ |
| $S_T \to 0$ | 趋近$-c_1$ | $c_2$ | 趋近$c_2 - c_1$ |

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# 看涨期权的正向差期组合盈亏

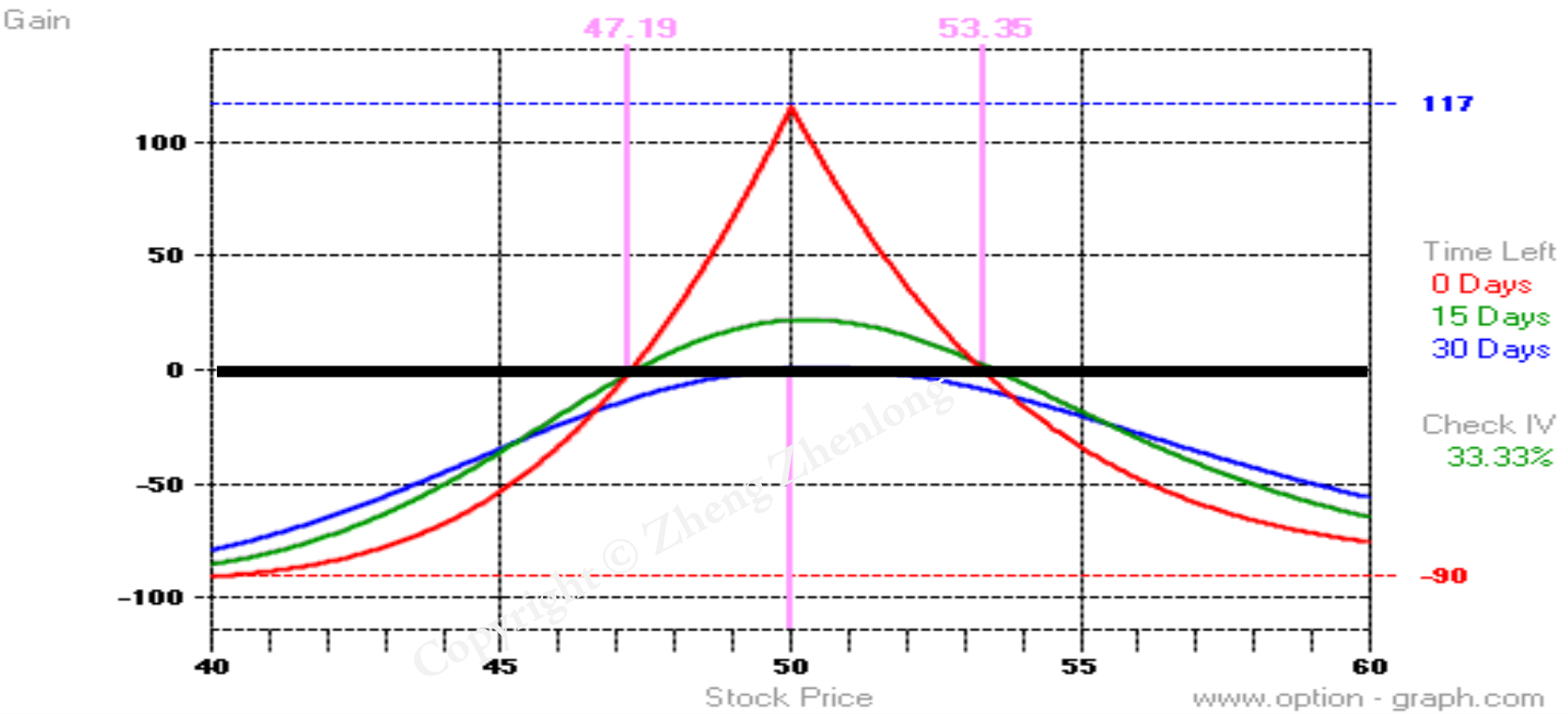



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**Position Details : Calendar Call, ATM**

| | | |
|---|---|---|
| Stock at 50, 1 Long Call, 50 strike, 65 days, IV 33.33%, Option Price $2.91 = | $291.00 | debit |
| Stock at 50, 1 Short Call, 50 strike, 30 days, IV 33.33%, Option Price $1.98 = | $198.00 | credit |

Total Position = $93.00 DEBIT



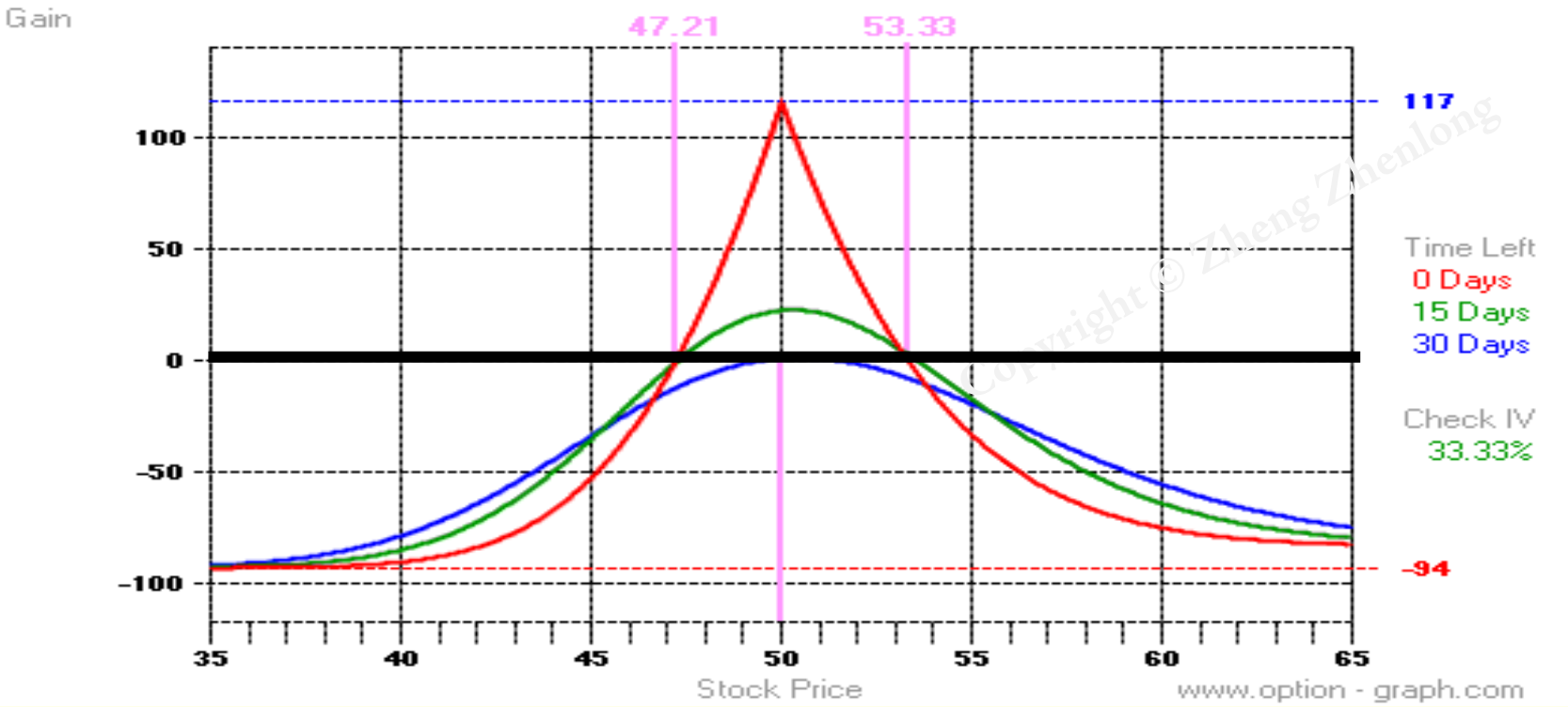

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## Position Details : Calendar Put, ATM

| | | |
|---|---|---|
| Stock at 50, 1 Long Put, 50 strike, 65 days, IV 33.33%, Option Price $2.73 = | $273.00 | debit |
| Stock at 50, 1 Short Put, 50 strike, 30 days, IV 33.33%, Option Price $1.89 = | $189.00 | credit |

Total Position = $84.00 DEBIT

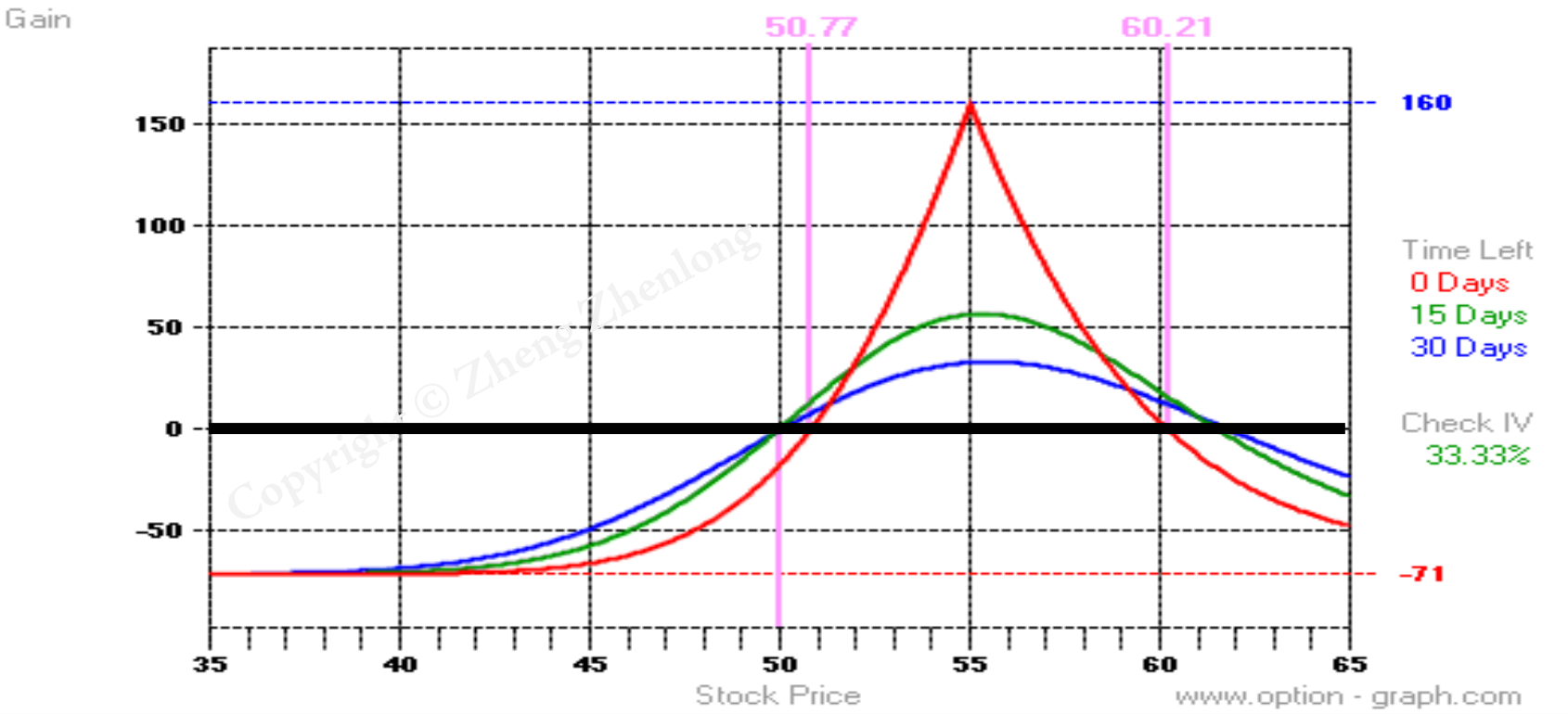



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## Position Details : Calendar Call, OTM

| | | |
|---|---|---|
| Stock at 50, 1 Long Call, 55 strike, 65 days, IV 33.33%, Option Price $1.16 = | $116.00 | debit |
| Stock at 50, 1 Short Call, 55 strike, 30 days, IV 33.33%, Option Price $0.45 = | $45.00 | credit |

Total Position = $71.00 DEBIT

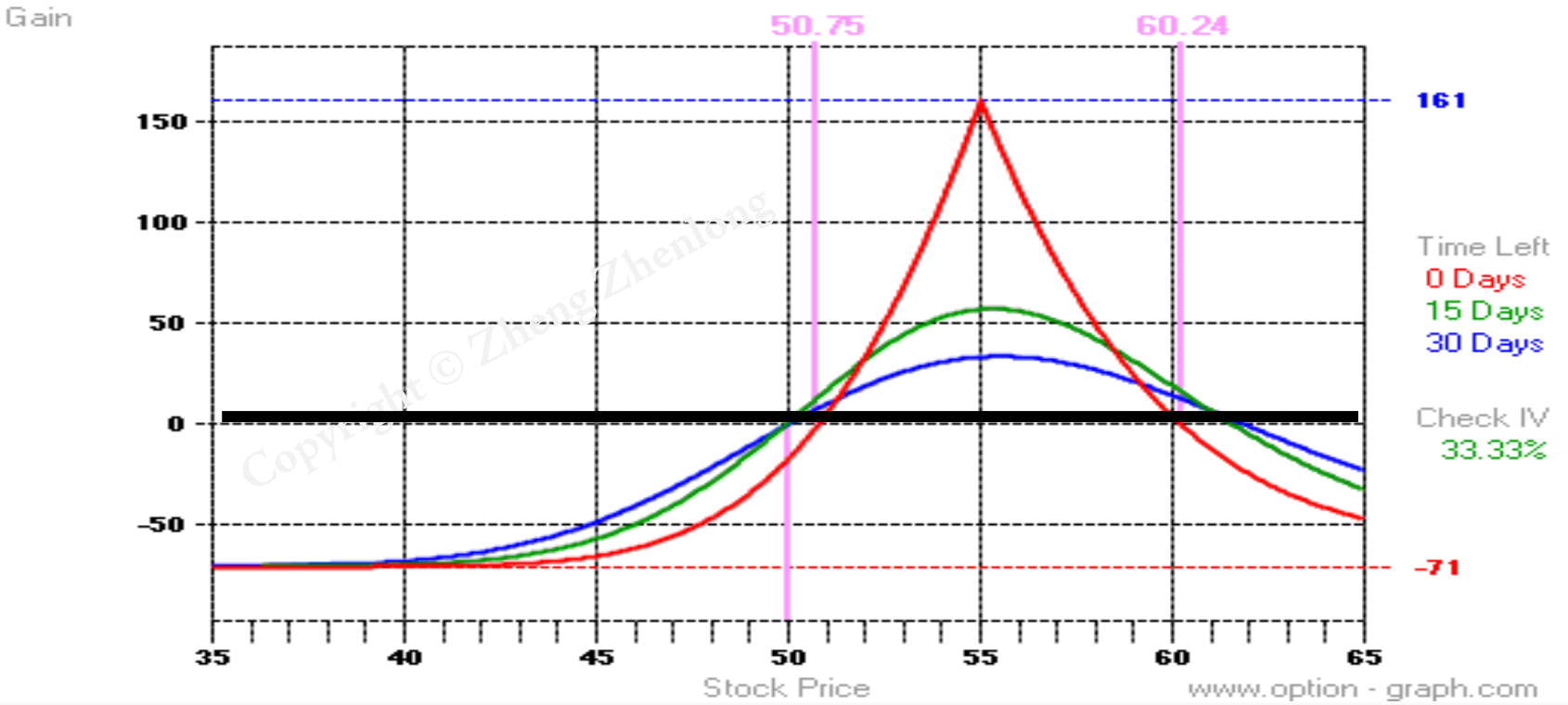



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## Position Details : Calendar Put, ITM

| | | |
|---|---|---|
| Stock at 50, 1 Long Put, 55 strike, 65 days, IV 33.33%, Option Price $5.96 = | $596.00 | debit |
| Stock at 50, 1 Short Put, 55 strike, 30 days, IV 33.33%, Option Price $5.36 = | $536.00 | credit |

Total Position = $60.00 DEBIT

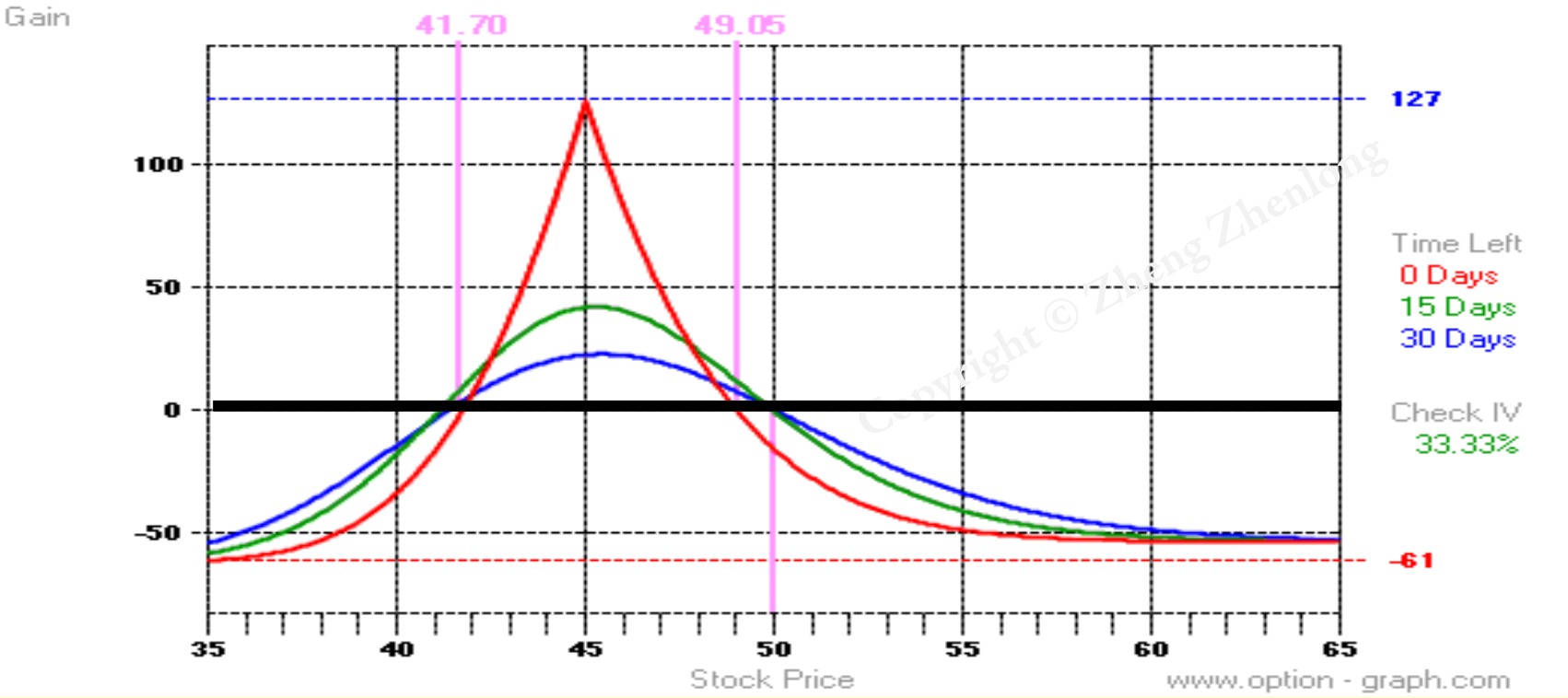



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## Position Details : Calendar Call, ITM

| | | |
|---|---|---|
| Stock at 50, 1 Long Call, 45 strike, 65 days, IV 33.33%, Option Price $6.01 = | $601.00 | debit |
| Stock at 50, 1 Short Call, 45 strike, 30 days, IV 33.33%, Option Price $5.39 = | $539.00 | credit |

Total Position = $62.00 DEBIT

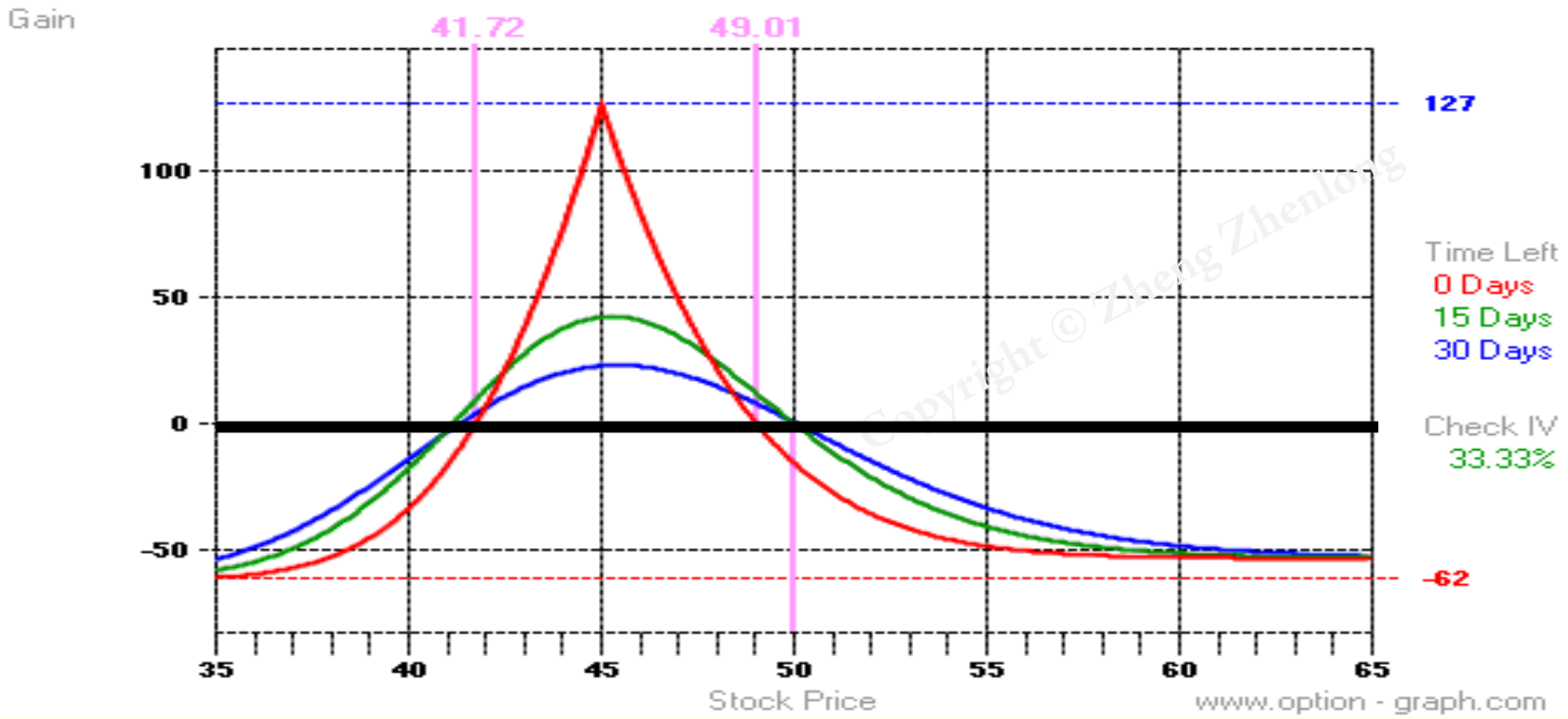



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## Position Details : Calendar Put, OTM

| | | |
|---|---|---|
| Stock at 50, 1 Long Put, 45 strike, 65 days, IV 33.33%, Option Price $0.85 = | $85.00 | debit |
| Stock at 50, 1 Short Put, 45 strike, 30 days, IV 33.33%, Option Price $0.31 = | $31.00 | credit |

Total Position = $54.00 DEBIT




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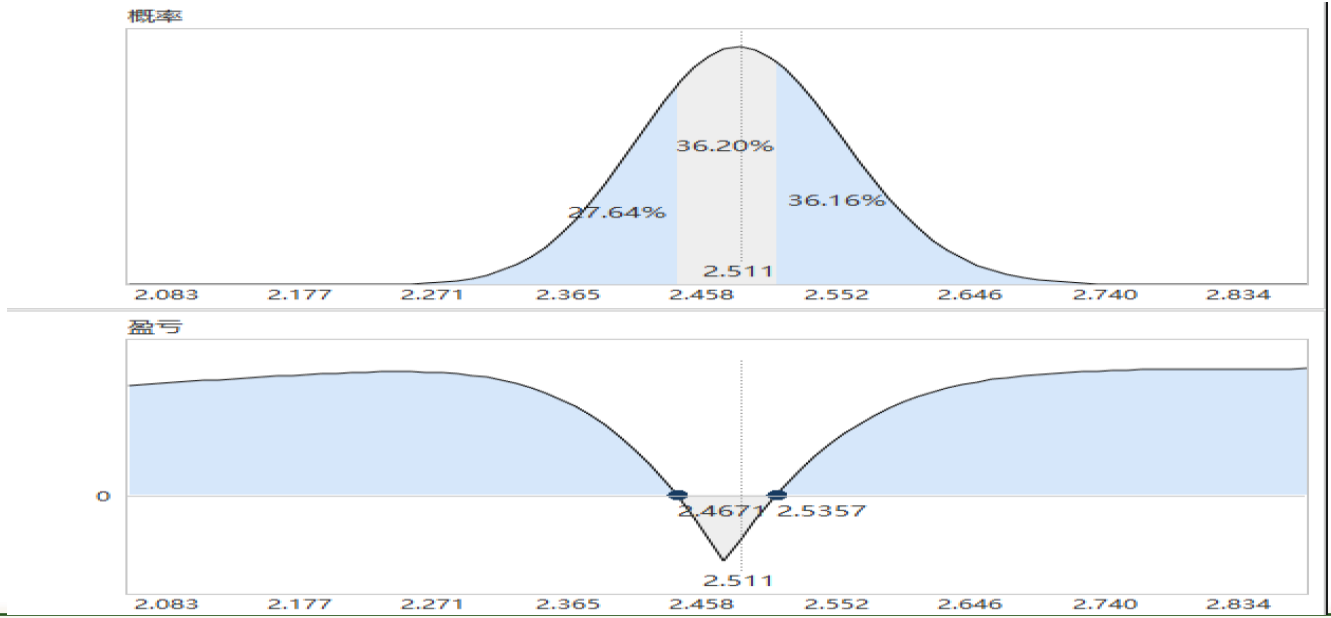
# 反向差期组合（19.2.15）

| 买卖 | 名称 | 手数 | 价格类型 | 价格 | 买价 | 卖价 | 隐含波动率 | Delta | Gamma | Vega | Theta | 合约乘数 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 卖 | 50ETF沽3月2.50A | 1 | 买价 | 0.0552 | 0.0552 | 0.0556 | 19.07% | -0.4362 | 3.0123 | 0.0033 | -0.0006 | 10202 |
| 买 | 50ETF沽2月2.50 | 1 | 卖价 | 0.0279 | 0.0278 | 0.0279 | 18.02% | -0.4236 | 5.3194 | 0.0019 | -0.0010 | 10000 |

标的　　保存策略

最大收益: +∞　最大亏损: -174.02

总Delta: 225.93　总Gamma: 23612.93　总Vega: -15.58　总Theta: -5.24




不对

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# 对角组合（Diagonal Spreads）

- 对角组合（Diagonal Spreads）
  - 两份行权价格不同（$K_1$ 和 $K_2$，且 $K_1 < K_2$）、期限也不同（T 和 T*，且 T < T*）的同种期权的不同头寸组成。
  - 它有八种类型
    - 看涨/看跌
    - 牛市/熊市——买低卖高/买高卖低
    - 正向/反向——买长卖短/买短卖长

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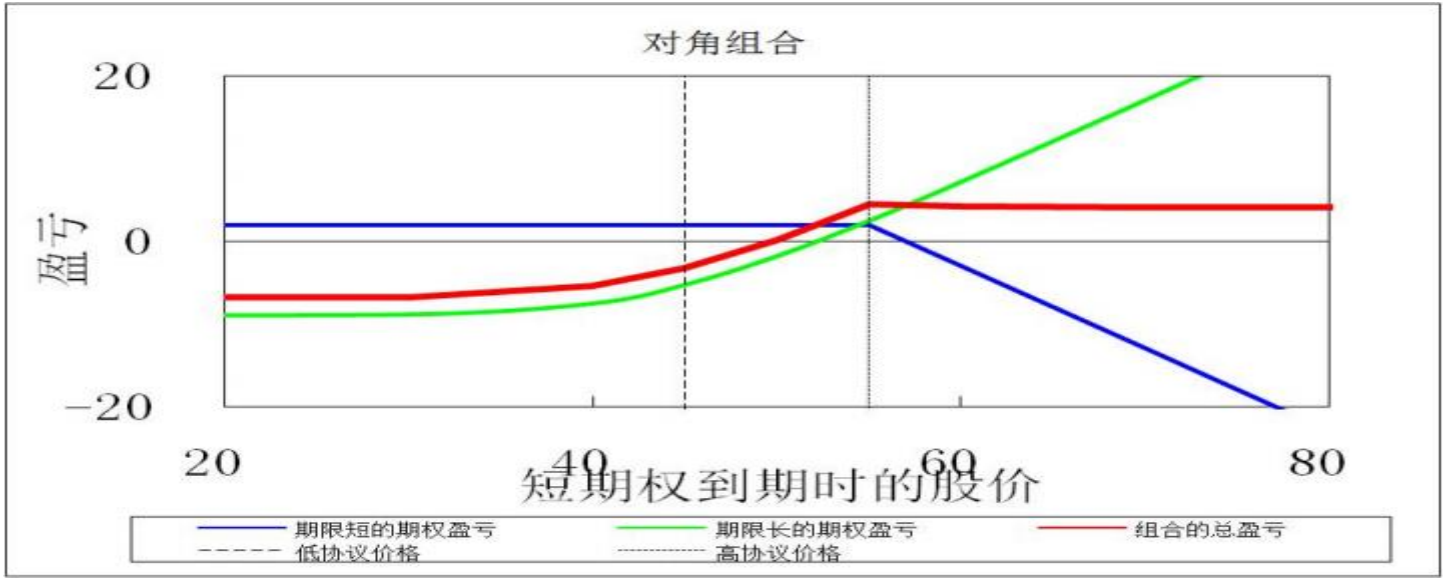
# 看涨期权的牛市正向对角组合

* 由看涨期权的（$K_1$,T*）多头加（$K_2$,T）空头组合组成

| $S_T$ 的范围 | $(X_1, T^*)$ 多头的盈亏 | $(X_2, T)$ 空头的盈亏 | 总盈亏 |
|---|---|---|---|
| $S_T \to \infty$ | 趋近 $S_T - X_1 e^{-r(T^*-T)} - c_1$ | $X_2 - S_T + c_2$ | 趋近 $X_2 - X_1 e^{-r(T^*-T)} + c_2 - c_1$ |
| $S_T = X_2$ | $X_2 - X_1 e^{-r(T^*-T)} + c_{1T} - c_1$ | $c_2$ | $X_2 - X_1 e^{-r(T^*-T)} + c_2 - c_1 + c_{1T}$ |
| $S_T \to 0$ | 趋近 $-c_1$ | $c_2$ | 趋近 $c_2 - c_1$ |

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# 看涨期权的牛市正向对角组合盈亏




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# 对角组合的八种类型(I)

- 看涨期权的牛市正向对角组合
  - 由看涨期权的（$K_1$，T*）多头加（$K_2$，T）空头组合组成
- 看涨期权的熊市反向对角组合
  - 由看涨期权的（$K_1$，T*）空头加（$K_2$，T）多头组成的组合。其盈亏图与1刚好相反
- 看涨期权的熊市正向对角组合
  - 由看涨期权的（$K_2$，T*）多头加（$K_1$，T）空头组成的组合。
- 看涨期权的牛市反向对角组合。
  - 由看涨期权的（$K_2$，T*）空头加（$K_1$，T）多头组成的组合，其盈亏图与3刚好相反。

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* 看跌期权的牛市正向对角组合。
  * 由看跌期权的（$K_1$，T*）多头加（$K_2$，T）空头组成的组合。
* 看跌期权的熊市反向对角组合。
  * 由看跌期权的（$K_1$，T*）空头加（$K_2$，T）多头组成的组合，其盈亏图与5刚好相反。
* 看跌期权的熊市正向对角组合。
  * 由看跌期权的（$K_2$，T*）多头加（$K_1$，T）空头组成的组合。
* 看跌期权的牛市反向对角组合。
  * 由看跌期权的（$K_2$，T*）空头加（$K_1$，T）多头组成的组合，其盈亏图与7刚好相反。

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# 牛市反向对角组合

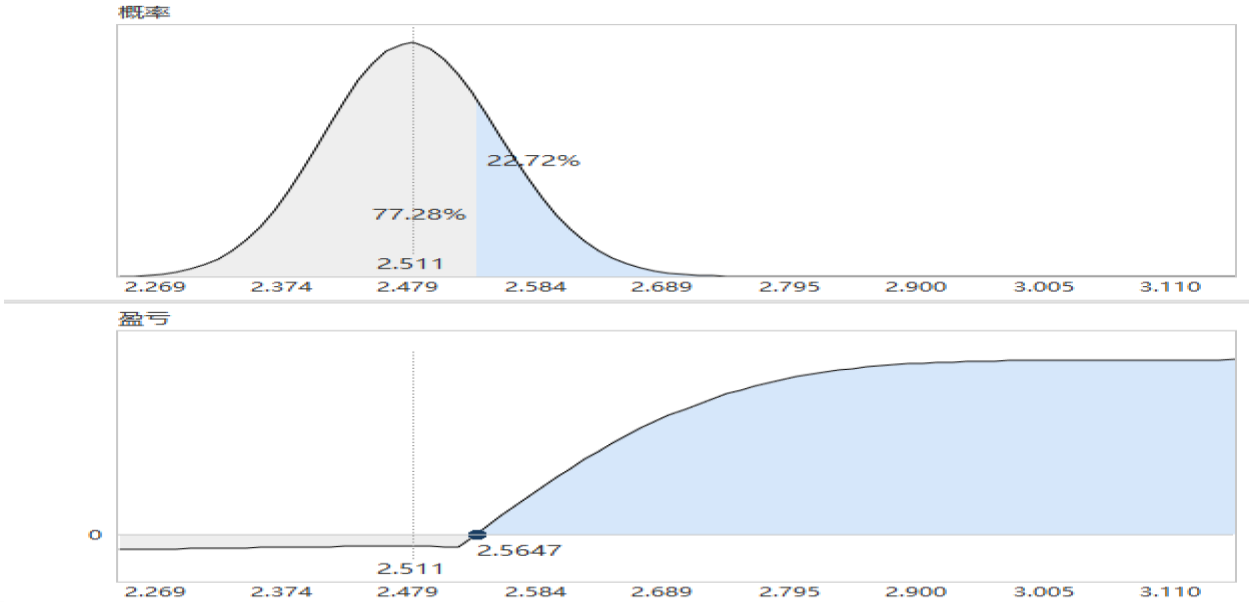
| 买卖 | 名称 | 手数 | 价格类型 | 价格 | 买价 | 卖价 | 隐含波动率 | Delta | Gamma | Vega | Theta | 合约乘数 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 买 | 50ETF沽2月2.55 | 1 | 卖价 | 0.0556 | 0.0555 | 0.0556 | 17.84% | -0.6854 | 4.8229 | 0.0017 | -0.0009 | 10000 |
| 卖 | 50ETF沽3月2.745A | 1 | 买价 | 0.2335 | 0.2335 | 0.2345 | 19.32% | -0.9490 | 0.8016 | 0.0009 | - | 10202 |

标的 | 保存策略

最大收益: +∞ 最大亏损: -626.91

总Delta: 2770.57 总Gamma: 40322.67 总Vega: 6.88 总Theta: -9.41

不对

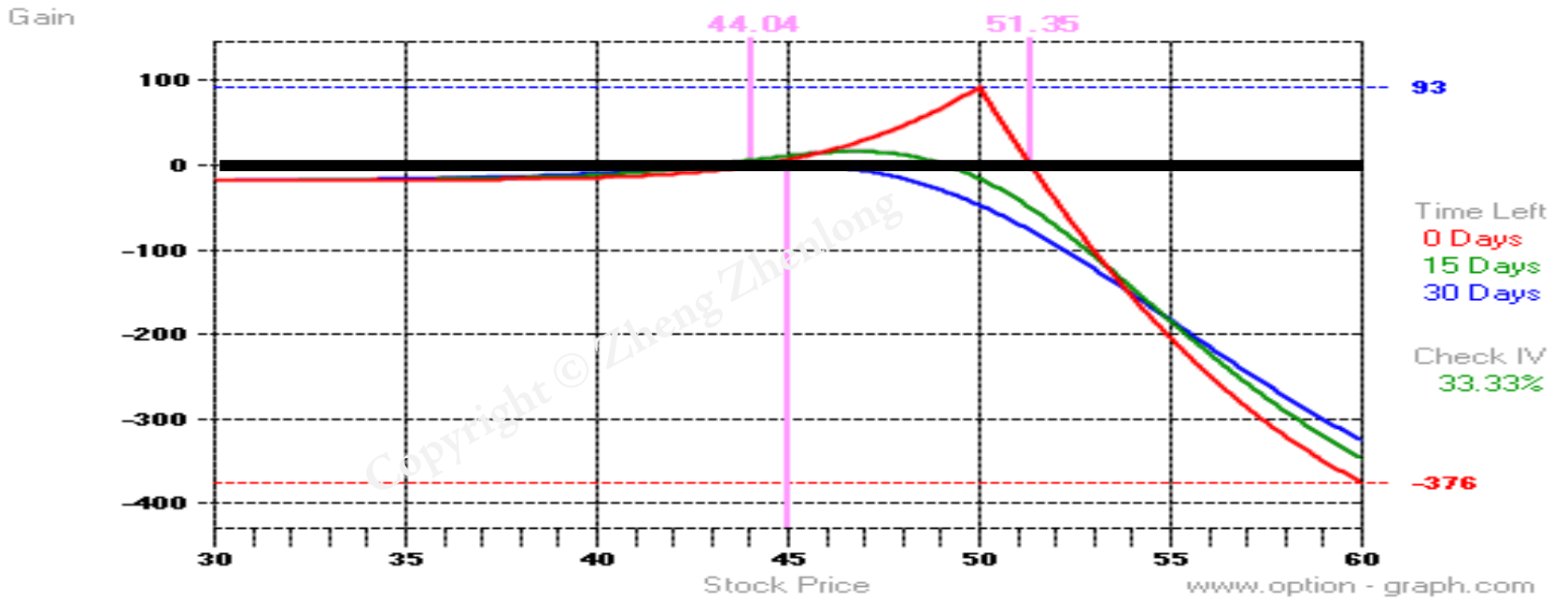



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# 看涨期权的熊市正向对角组合 Profit

**Position Details : Bear Call Diagonalized by 2 months**

| | | |
|---|---|---|
| Stock at $45.00, 1 Long Call, 55 strike, 93 days, IV 33.33%, Option Price $0.51 = | $51.00 | debit |
| Stock at $45.00, 1 Short Call, 50 strike, 30 days, IV 33.33%, Option Price $0.34 = | $34.00 | credit |
| Total Position = $17.00 DEBIT | | |




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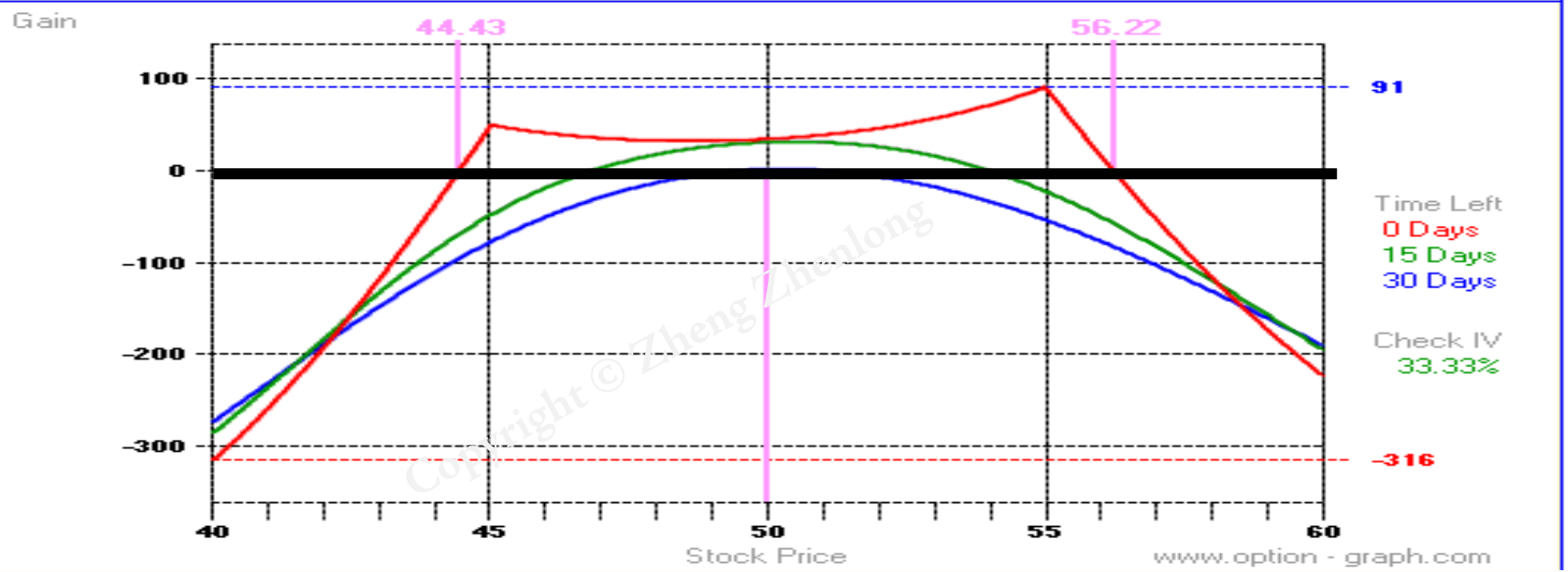
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# 双对角组合：空头行权价格不同

**Position Details : Double Diagonal with short strikes different**

| | | |
|---|---|---|
| Stock at 50, 1 Short Call, 55 strike, 30 days, IV 33.33%, Option Price $0.45 = | $45.00 | credit |
| Stock at 50, 1 Short Put, 45 strike, 30 days, IV 33.33%, Option Price $0.31 = | $31.00 | credit |
| Stock at 50, 1 Long Call, 60 strike, 65 days, IV 33.33%, Option Price $0.38 = | $38.00 | debit |
| Stock at 50, 1 Long Put, 40 strike, 65 days, IV 33.33%, Option Price $0.15 = | $15.00 | debit |

Total Position = $23.00 CREDIT



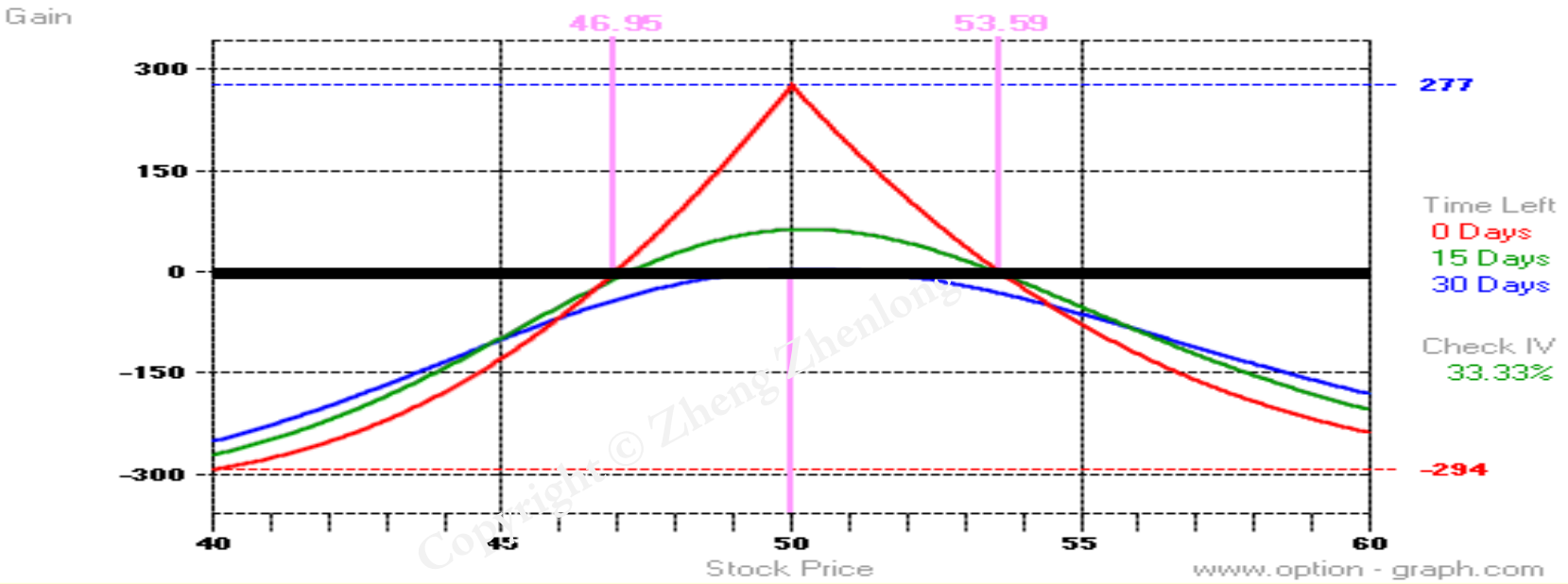

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# 双对角组合：空头行权价格相同

**Position Details : Double Diagonal using same short strike**

| | | |
|---|---|---|
| Stock at 50, 1 Short Call, 50 strike, 30 days, IV 33.33%, Option Price $1.98 = | $198.00 | credit |
| Stock at 50, 1 Short Put, 50 strike, 30 days, IV 33.33%, Option Price $1.89 = | $189.00 | credit |
| Stock at 50, 1 Long Call, 55 strike, 65 days, IV 33.33%, Option Price $1.16 = | $116.00 | debit |
| Stock at 50, 1 Long Put, 45 strike, 65 days, IV 33.33%, Option Price $0.85 = | $85.00 | debit |

Total Position = $186.00 CREDIT



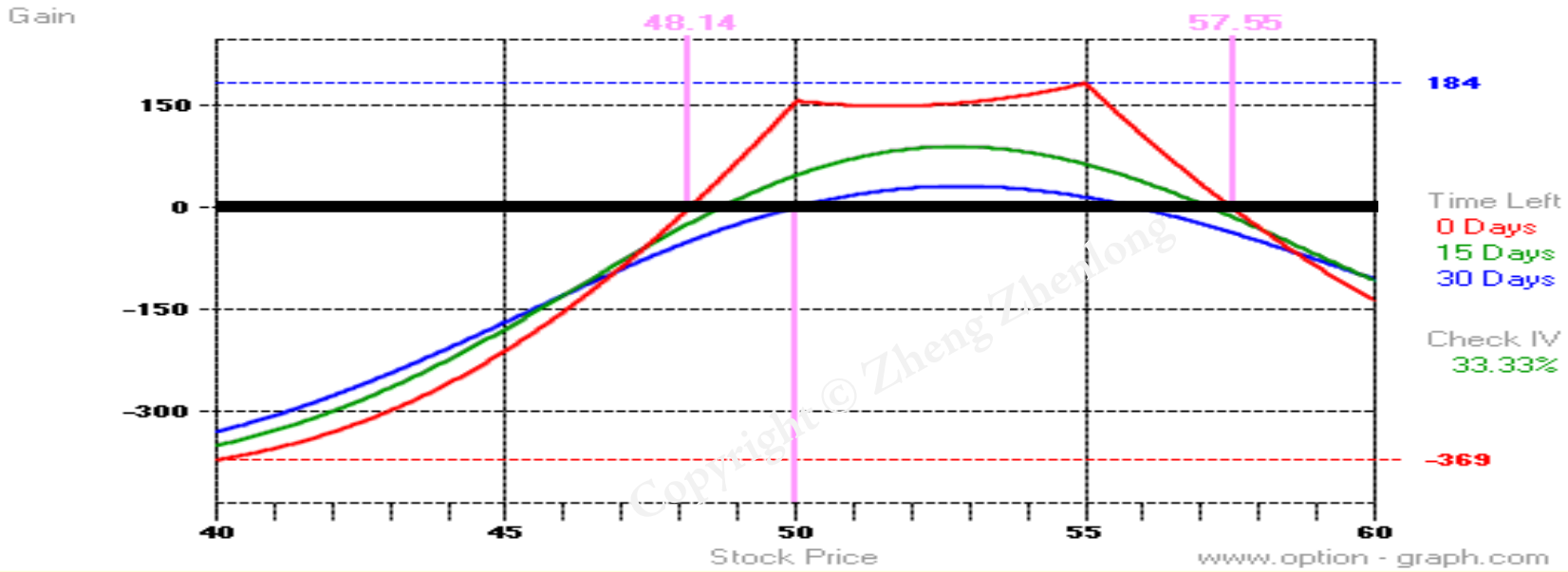

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# 双对角组合：牛市正向

**Position Details : Double Diagonal, bullish**

| | | |
|---|---|---|
| Stock at 50, 1 Short Call, 55 strike, 30 days, IV 33.33%, Option Price $0.45 = | $45.00 | credit |
| Stock at 50, 1 Short Put, 50 strike, 30 days, IV 33.33%, Option Price $1.89 = | $189.00 | credit |
| Stock at 50, 1 Long Call, 60 strike, 65 days, IV 33.33%, Option Price $0.38 = | $38.00 | debit |
| Stock at 50, 1 Long Put, 45 strike, 65 days, IV 33.33%, Option Price $0.85 = | $85.00 | debit |

Total Position = $111.00 CREDIT



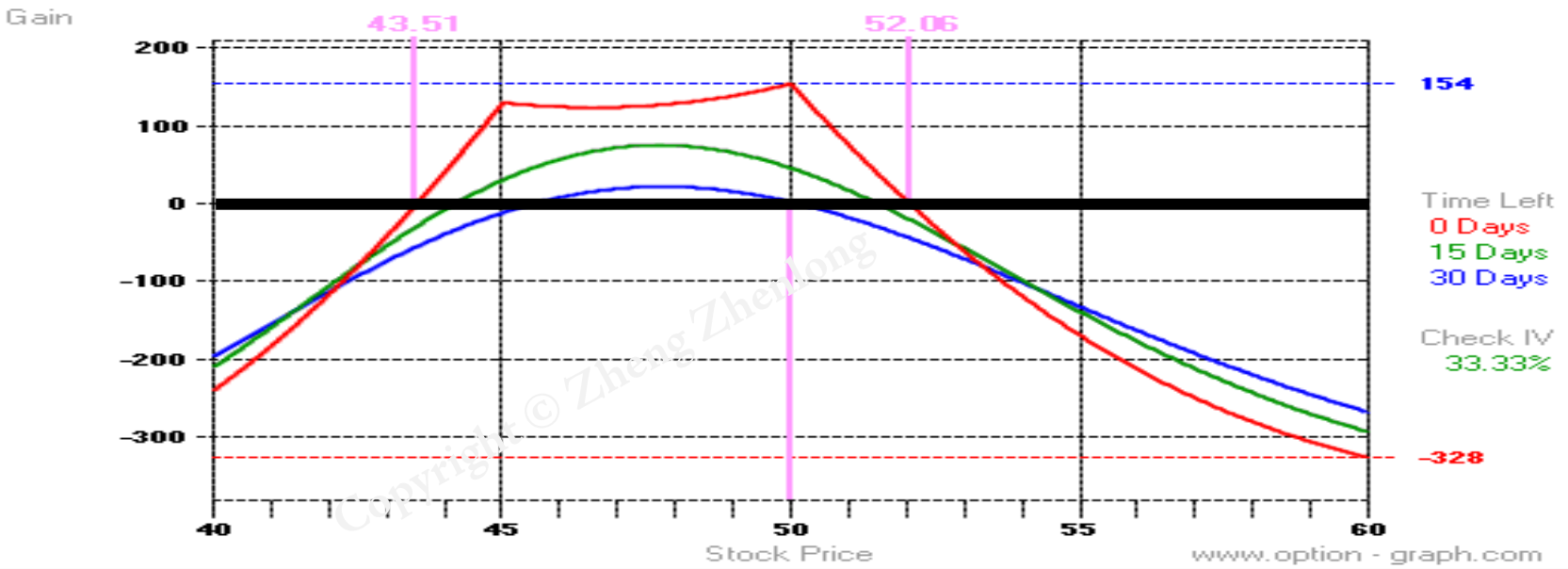

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# 双对角组合：熊市正向

**Position Details : Double Diagonal, bearish**

| | | |
|---|---|---|
| Stock at 50, 1 Short Call, 50 strike, 30 days, IV 33.33%, Option Price $1.98 = | $198.00 | credit |
| Stock at 50, 1 Short Put, 45 strike, 30 days, IV 33.33%, Option Price $0.31 = | $31.00 | credit |
| Stock at 50, 1 Long Call, 55 strike, 65 days, IV 33.33%, Option Price $1.16 = | $116.00 | debit |
| Stock at 50, 1 Long Put, 40 strike, 65 days, IV 33.33%, Option Price $0.15 = | $15.00 | debit |

Total Position = $98.00 CREDIT




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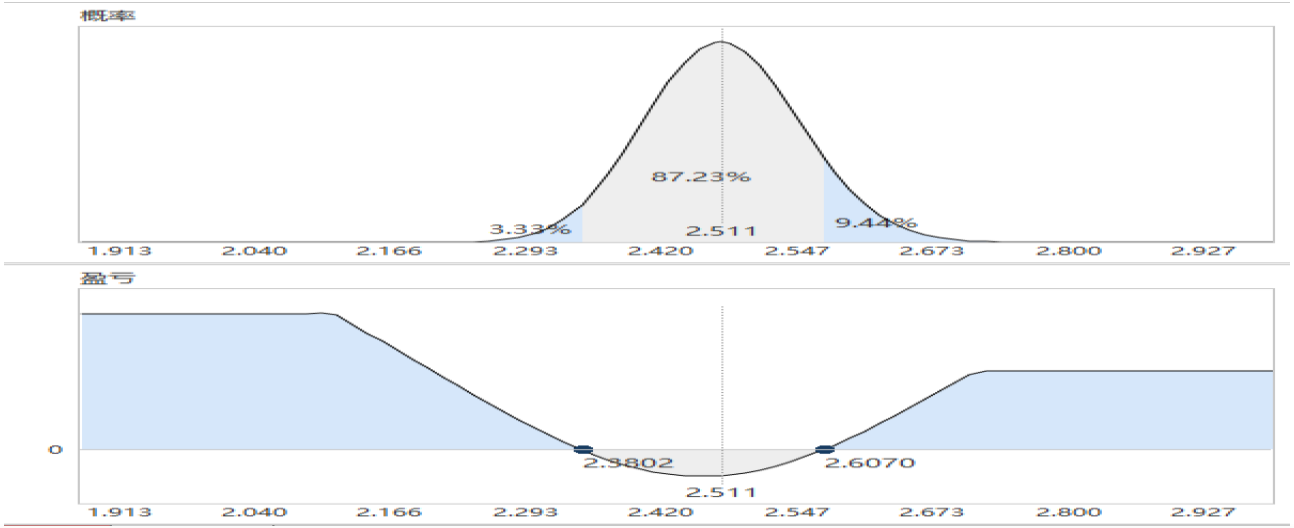
# 双对角组合（19.2.15）

| 买卖 | 名称 | 手数 | 价格类型 | 价格 | 买价 | 卖价 | 隐含波动率 | Delta | Gamma | Vega | Theta | 合约乘数 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 卖 | 50ETF购2月2.15 | 1 | 买价 | 0.3632 | 0.3632 | 0.3642 | 38.28% | 1.0000 | - | - | -0.0001 | 10000 |
| 卖 | 50ETF沽2月2.75 | 1 | 买价 | 0.2369 | 0.2369 | 0.2371 | 19.92% | -0.9989 | 0.0505 | - | 0.0002 | 10000 |
| 买 | 50ETF购3月2.50 | 1 | 卖价 | 0.0766 | 0.0765 | 0.0766 | 20.15% | 0.5638 | 3.0123 | 0.0033 | -0.0007 | 10000 |
| 买 | 50ETF沽3月2.50 | 1 | 卖价 | 0.0553 | 0.0552 | 0.0553 | 19.02% | -0.4362 | 3.0123 | 0.0033 | -0.0006 | 10000 |

标的　　保存策略

最大收益: 2136.35　最大亏损: -∞

总Delta: 1242.55　总Gamma: 59005.02　总Vega: 65.36　总Theta: -13.47




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# 混合期权

- 混合期权
  - 由不同种的期权——看涨和看跌组成的组合
  - 行权价格、期限、多空、份数等可以相同或不同
  - 底部/顶部——多头/空头

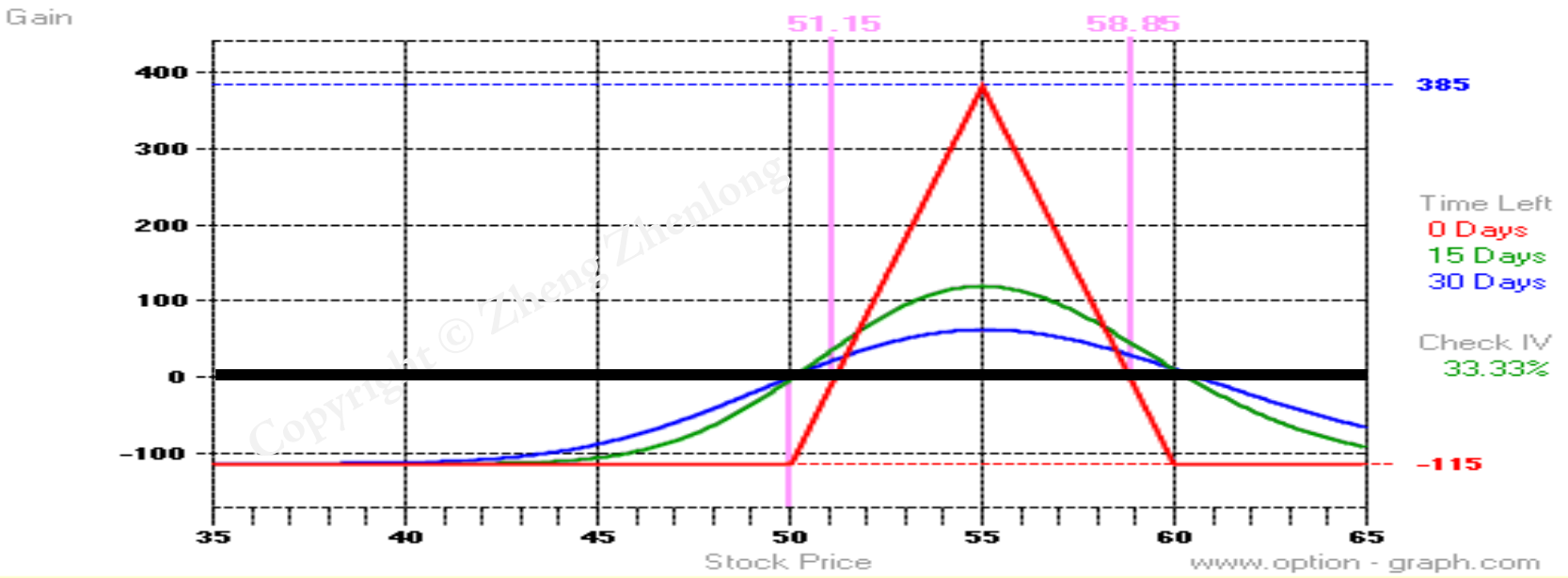
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# 正向蝶式差价组合：认购＋认沽

**Position Details : Butterfly, Bullish**

| | | |
|---|---|---|
| Stock at 50, 1 Short Call, 55 strike, 30 days, IV 33.33%, Option Price $0.45 = | $45.00 | credit |
| Stock at 50, 1 Short Put, 55 strike, 30 days, IV 33.33%, Option Price $5.36 = | $536.00 | credit |
| Stock at 50, 1 Long Call, 60 strike, 30 days, IV 33.33%, Option Price $0.07 = | $7.00 | debit |
| Stock at 50, 1 Long Put, 50 strike, 30 days, IV 33.33%, Option Price $1.89 = | $189.00 | debit |

Total Position = $385.00 CREDIT



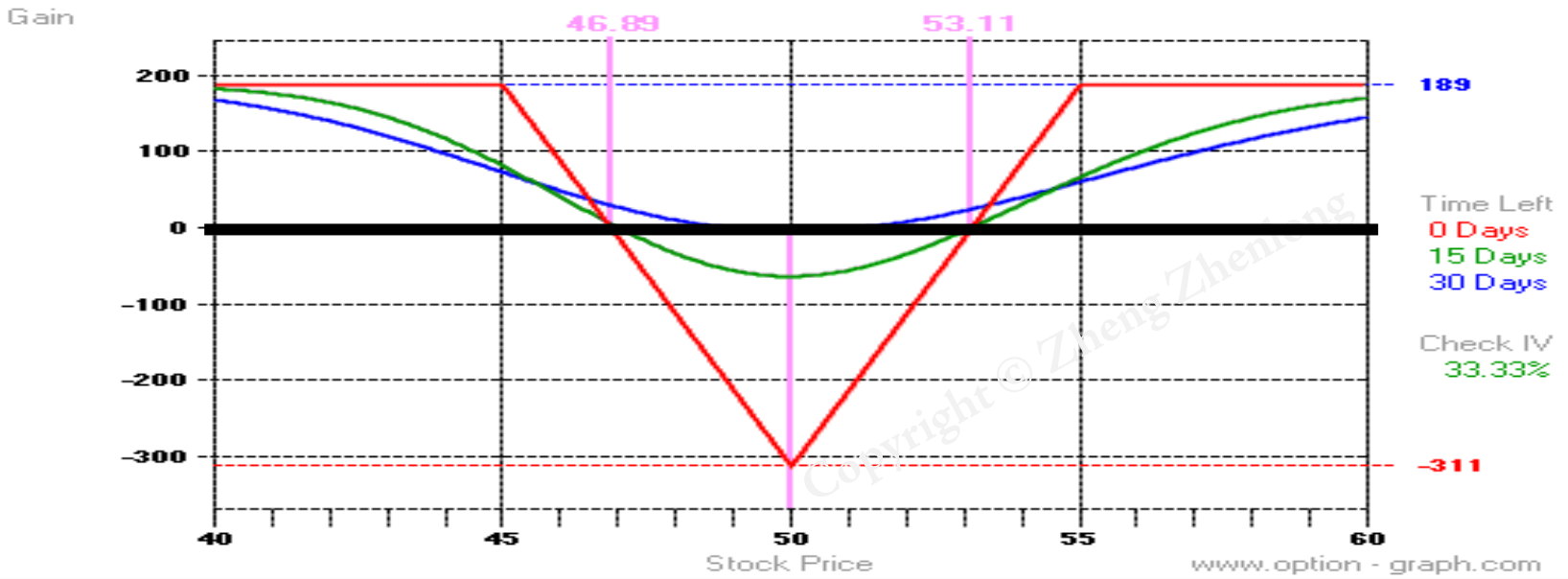

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# 反向蝶式差价组合：认购＋认沽

**Position Details : Butterfly, Reverse**

| | | |
|---|---|---|
| Stock at 50, 1 Long Call, 50 strike, 30 days, IV 33.33%, Option Price $1.98 = | $198.00 | debit |
| Stock at 50, 1 Long Put, 50 strike, 30 days, IV 33.33%, Option Price $1.89 = | $189.00 | debit |
| Stock at 50, 1 Short Call, 55 strike, 30 days, IV 33.33%, Option Price $0.45 = | $45.00 | credit |
| Stock at 50, 1 Short Put, 45 strike, 30 days, IV 33.33%, Option Price $0.31 = | $31.00 | credit |

Total Position = $311.00 DEBIT



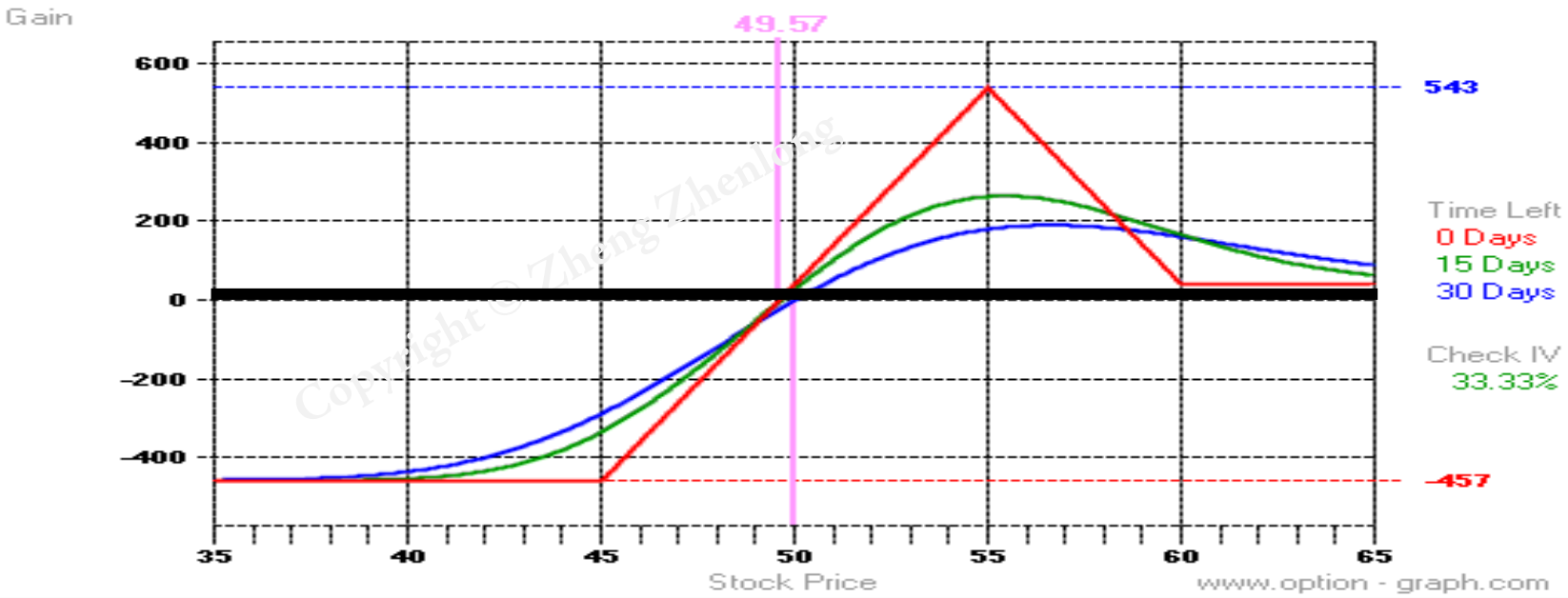

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# 牛市蝶式差价：认购＋认沽

**Position Details : Butterfly, Modified Bullish**

| | | |
|---|---|---|
| Stock at 50, 1 Short Call, 55 strike, 30 days, IV 33.33%, Option Price $0.45 = | $45.00 | credit |
| Stock at 50, 1 Short Put, 55 strike, 30 days, IV 33.33%, Option Price $5.36 = | $536.00 | credit |
| Stock at 50, 1 Long Call, 60 strike, 30 days, IV 33.33%, Option Price $0.07 = | $7.00 | debit |
| Stock at 50, 1 Long Put, 45 strike, 30 days, IV 33.33%, Option Price $0.31 = | $31.00 | debit |

Total Position = $543.00 CREDIT




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# 跨式组合（Straddle）

* 跨式组合（Straddle）
  * 由具有相同行权价格、相同期限的一份看涨期权和一份看跌期权组成
  * 底部跨式组合：两份多头组成
  * 顶部跨式组合：两份空头组成

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# 条式组合（Strip）

* 条式组合（Strip）
  * 由具有相同行权价格、相同期限的一份看涨期权和两份看跌期权组成
  * 条式组合也分底部和顶部两种，前者由多头构成，后者由空头构成
  * 具有不对称性，底部条式适合投资者预测价格变化较大，且下跌的可能大于上涨可能的情形

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# 带式组合（Strap）

* 带式组合（Strap）
  * 由具有相同行权价格、相同期限的资产的两份看涨期权和一份看跌期权组成
  * 带式组合也分底部和顶部两种，前者由多头构成，后者由空头构成
  * 具有不对称性，适应于投资者预测价格变化较大，且上升的可能大于下跌可能的情形

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# 宽跨式组合（Strangle）

- 宽跨式组合（Strangle）
  - 由相同到期日但行权价格不同的一份看涨期权和一份看跌期权组成，其中看涨期权的行权价格高于看跌期权
  - 宽跨式组合也分底部和顶部，前者由多头组成，后者由空头组成

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# 混合期权的比较

- 此类策略适合预期标的资产价格有大幅波动但不能确定方向的投资者
- 跨式组合是对称的，条式和带式分别适合有一定预期偏好的投资者
- 相比跨式组合，宽跨式组合只有在股价有更大幅度波动时才可能获利，但其初始成本也相对较低，其收益特征取决于两个执行价格的接近程度

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# 目录

期权交易头寸及其应用

期权交易策略及其应用

期权组合盈亏图的算法

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# 期权组合盈亏图的算法

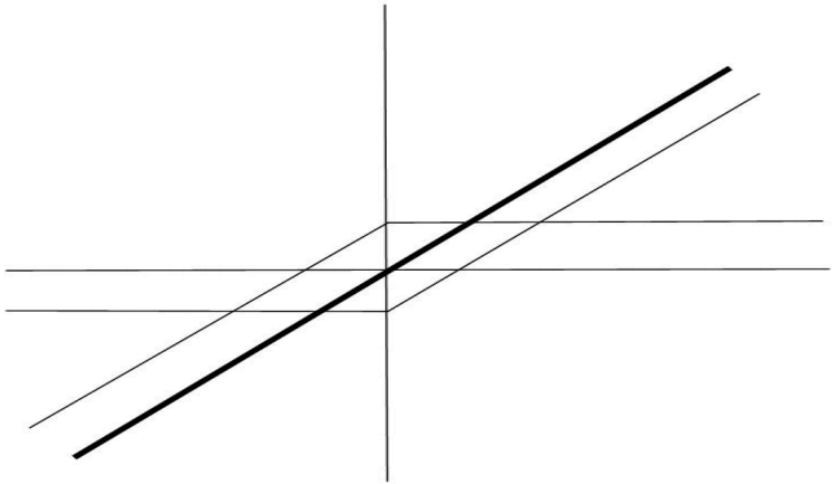
* 期权组合的盈亏可以通过符号的方法形象化的表示
* 符号规则：
  * 期权交易的结果在盈亏图上出现负斜率：（–1）
  * 期权交易的结果在盈亏图上出现正斜率：（+1）
  * 期权交易的结果在盈亏图上是水平状：（0）

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# 标的资产多头的组合分解图

* 看涨多头＋看跌空头＝标的资产多头：

$$(0,+1)+(+1,0)=(+1,+1)$$

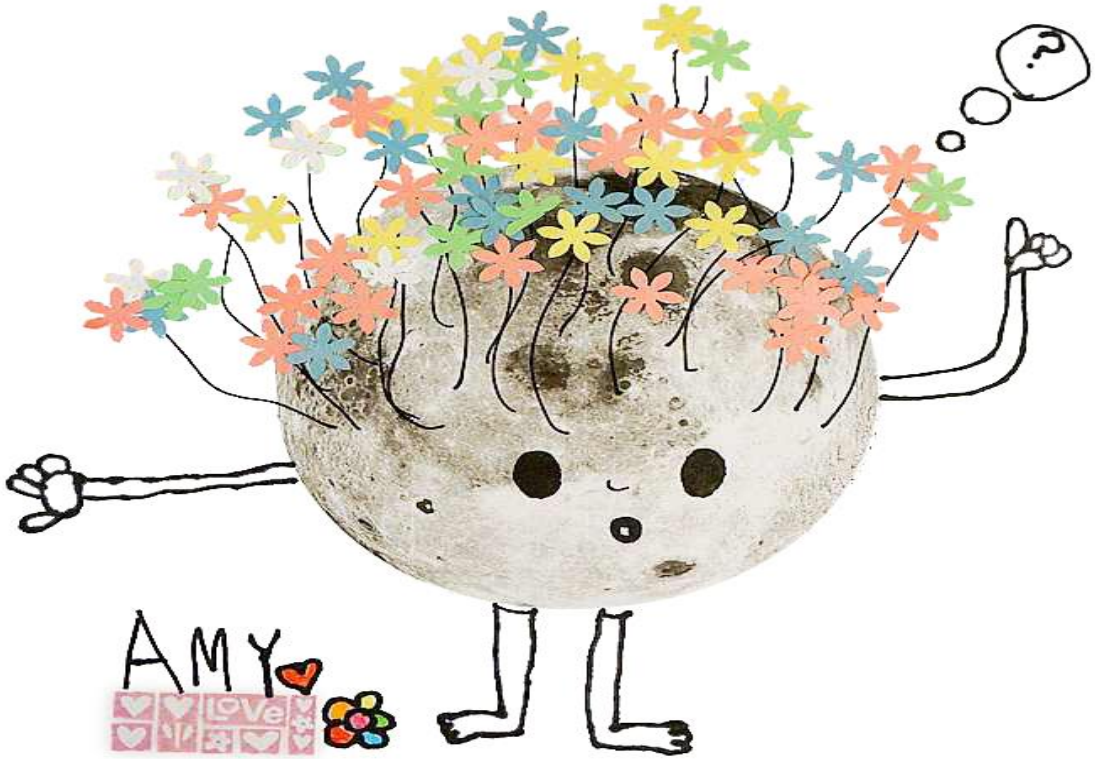


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* 各种基本头寸的盈亏状态
  * 看涨多头：$(0, +1)$
  * 看涨空头：$(0, -1)$
  * 看跌多头：$(-1, 0)$
  * 看跌空头：$(+1, 0)$
  * 标的资产多头：$(+1, +1)$
  * 标的资产空头：$(-1, -1)$

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# Any Questions ?




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